Fitbauer

User manual

Software for Mössbauer spectrum fitting and analysis

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Abstract

Fitbauer is a desktop application to load, fold, *simulate* and *fit* ^{57}Fe Mössbauer spectra. The simulation corresponds to the *forward model* —singlets, doublets and sextets—, while the fit can be *discrete* (those same components) or via *distributions* of hyperfine field or quadrupole splitting, which are obtained by **fitting** the data (a probability distribution is inferred from the spectrum, not arbitrarily "simulated").

Fitbauer was born with one idea: to build something **new, easy to understand and intuitive**, with **live simulation** and everything needed to fit Mössbauer spectra without friction. Other programs exist —some commercial, some free or academic— but they did not always show or offer what was needed or wanted for day-to-day work; hence this project. Section 1.2 explains why, and section 1.3 places it, neutrally, alongside other tools in the field.

The manual follows the real workflow: first *calibration* —which is fixed and reused—, then loading and visualising data, simulating and fitting a calibrant, and finally distribution fitting. The **Note**, **Important** and **Tip** boxes highlight practical details.

Contents

Chapter 1

Introduction and getting started

1.1 What is Fitbauer

Fitbauer is a desktop application for the analysis of ^{57}Fe Mössbauer spectra. It lets you:

- **Load** spectra in several formats (raw counts `.ws5`/`.adt` or spectra already folded in velocity `.csv`/`.txt`/`.dat`/`.exp`).
- **Fold** the raw counts and build the velocity axis from a *calibration*.
- **Simulate** the *forward model*, live: up to six components (singlet, doublet, sextet and magnetic relaxation models), with Lorentz/Voigt profiles.
- **Fit** the data, either *discretely* (those same components) or through *distributions* $P(B_{\mathrm{hf}})$ / $P(\Delta E_Q)$.
- **Document**: Markdown/PDF reports, figure export and reproducible sessions.

The interface is translated into seven languages (**Language** menu): Spanish, English, French, German, simplified Chinese, Japanese and Russian.

> **Note**
>
> **Simulating** and **fitting** are not the same thing. *Simulation* generates a spectrum from parameters that you set (forward model). *Fitting* does the opposite: it starts from the measured spectrum and *infers* the parameters that best reproduce it. This is why a *probability distribution* is not "simulated" out of thin air: it is **obtained by fitting** the spectrum (although you can *simulate the spectrum* that a given distribution would produce).

> **Note**
>
> The computation (physics and fitting engines) resides in the `core/` package, which does not depend on any graphical interface. The GUI is a client of that core. This manual describes how to use the application; the mathematical detail of the fitting engine is in **appendix** A (*Mathematical manual of the fitting engine*) and in the other mathematical appendices.

1.2 Why Fitbauer?

Fitbauer grew out of a practical need in the laboratory. The idea was to build something **new, easy to understand and intuitive**, that put **live simulation** first —seeing immediately how each parameter deforms the spectrum— and that included **what is necessary** to fit Mössbauer spectra without unnecessary learning curves.

Other programs exist for this purpose, some commercial and others free or for academic use. They are valuable, well-established tools, but in day-to-day work they did not always show or offer what was needed or wanted: sometimes because they rely on old environments, because of non-intuitive interfaces, because they were not available on every platform, or simply because they did not fit the specific workflow of our group.

With that motivation, Fitbauer pursues a few principles:

- **Immediacy**: load, view and simulate in a few clicks; the plot responds live to every parameter change.
- **Clarity**: parameters with physical meaning, explicit units and a calibration that is set once and reused.
- **Reproducibility**: sessions and reports that let you redo an analysis exactly, and a computation core kept separate from the interface.
- **Accessibility**: cross-platform, multilingual and easy to install (or as an executable, without installing Python).

1.3 Comparison with other programs

By way of orientation —and **without any intention of deciding which is best**—, it is useful to place Fitbauer alongside other tools common in the field. Each has its own audience, history and strengths:

- **NORMOS** (R. A. Brand): one of the historical standards, with site modules (*Site*) and distribution modules (*Dist*). Very widespread, traditionally tied to a DOS-like environment.
- **MossWinn** (Z. Klencsár): a very complete Windows program, with an extensive database and many models; commercial in nature.
- **Recoil** (Lagarec & Rancourt): popularized Voigt-based fitting (VBF) for distributions; oriented towards Windows.
- **Vinda** (H. P. Gunnlaugsson): a set of spreadsheet macros, free, widely used in environments such as ISOLDE.
- **WMOSS** and **Moessfit**: Windows alternatives with general fitting (including Hamiltonian diagonalization), for academic use.

Fitbauer sits in the territory of a **modern, cross-platform application focused on interactive simulation**, with the discrete and distribution fitting methods described in this manual. It does not aim to replace those above, but rather to offer a direct and convenient path for everyday work with Fe-57.

Note

The descriptions above are approximate and may have changed depending on the version of each program; they are included only to orient anyone coming from another tool.

1.4 Mössbauer spectroscopy at a glance

To follow the manual it is enough to remember the three hyperfine parameters that Fitbauer fits, and how they appear in the spectrum (velocity in mm/s versus transmission):

- **Isomer shift** δ: shifts the whole pattern along the velocity axis. It reports on the oxidation state and the coordination environment of the iron.
- **Quadrupole splitting** ΔE_Q: separates the lines into pairs. In a *doublet* it is the distance between its two peaks; it reflects the asymmetry of the electric field gradient.
- **Hyperfine field** B_{hf}: characteristic of magnetic materials. It generates a *sextet* of six lines whose separation grows with the field (for α-Fe, $B_{\mathrm{hf}} = 33.0$ T).

Depending on the material, the spectrum is a **singlet** (one line), a **doublet** (two), a **sextet** (six) or a **sum** of several of these contributions. When the iron environment is not unique but varies —disordered oxides, nanoparticles, alloys—, the spectrum is better described with a **distribution** of fields or quadrupoles (chapter 5).

1.5 Installation and startup

Fitbauer requires Python 3.11 or higher. From the repository root:

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
python fitbauer.py                 # abre la interfaz gráfica
```

The single entry point is `fitbauer.py`, which launches the main window (Qt + Matplotlib). Packaged executable versions (PyInstaller) are also available for those who do not wish to install Python.

1.6 Overview of the interface

On startup the main window appears, organized into:

- **Menu bar**: **File**, **Fit**, **View**, **Language** and **Help**.
- **Calibration panel**: control of $Vmax$, folding center, baseline, slope and line profile.
- **Component panels** (up to six): component shape and its physical parameters (δ, ΔE_Q, B_{hf}, linewidths, intensities).
- **Distribution panel**: $P(B_{\mathrm{hf}})/P(\Delta E_Q)$ mode, shape, regularizer and its parameters.
- **Plot area**: display of the spectrum, the model and the subspectra, with a navigation toolbar (Matplotlib).
- **Information/status panel**: fitted parameters, statistics (χ^2, AIC, BIC) and messages.

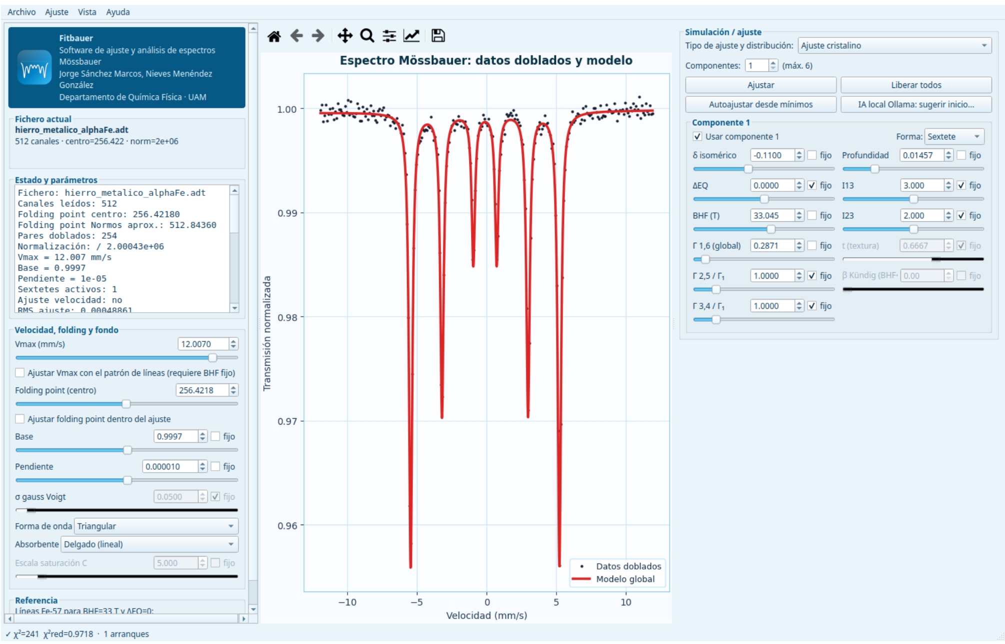

Figure 1.1: General view of the main Fitbauer window with its panels: menus, left-hand calibration and information panel, central plot area (here an α-Fe fit) and right-hand simulation/fitting panel.

Tip

The panel layout is configurable from **View > Configure panel layout...** You can save presets suited to large screens or laptops.

1.7 The typical workflow

The recommended order, which structures the following chapters, is:

1. **Calibrate** (chapter 2): load or define a velocity calibration and **set it**. From that moment on it is reused automatically.
2. **Load data** (chapter 3): open the target spectrum; the fixed calibration is applied by itself.
3. **Simulate and fit** (chapter 4): build the forward model and fit it; validate with an α-Fe calibration.
4. **Distributions** (chapter 5): when the hyperfine field is not unique, fit $P(B_{\mathrm{hf}})$ (and simulate the spectrum that a given distribution produces).

Chapter 2

Calibration

Calibration is **the first step** of the analysis. It determines the relationship between channel and velocity (mm/s) and, optionally, the isomer shift reference. In Fitbauer the calibration is **fixed** and reused in all subsequent loads, so it does not have to be redefined for each spectrum.

2.1 Why calibration is needed

The velocity transducer runs through a triangular cycle; the system records the counts per channel. To convert the channel axis into a physical velocity axis you need the **maximum velocity** $Vmax$ (mm/s at the end of the travel). This $Vmax$ depends on the transducer amplitude and is common to all spectra measured with the same configuration: this is why it is measured once with a standard and reused.

The usual standard is **metallic α-Fe**, whose sextet has very well-known line positions. Fitbauer adopts the convention that an α-Fe spectrum corresponds to $B_{\text{hf}} = 33{,}0$ T, using the published velocity standard ($\pm 0{,}839 / \pm 3{,}084 / \pm 5{,}329$ mm/s), just like NORMOS.

Important

The reference sextet positions are **not** derived from the nuclear moments: the textbook calculation gives a splitting $\sim 0{,}4\%$ smaller and would make the B_{hf} come out $\sim 0{,}1$ T too high. The published α-Fe positions at 33.0 T are used instead.

2.2 Defining a calibration from an α-Fe

The usual procedure is to load an α-Fe spectrum and use it to **fix the velocity scale**. It is worth keeping one idea clear:

Important

Calibrating means fitting the velocity, not the field. In α-Fe the hyperfine field **is already known**: by convention $B_{\text{hf}} = 33{,}0$ T. What is *not* known is the $Vmax$ —how many mm/s correspond to the end of the transducer travel—. That is why the calibration fit **fixes** $B_{\text{hf}} = 33{,}0$ T (and δ, ΔE_Q of α-Fe) and **leaves the $\boldsymbol{Vmax}$ free**, searching for the value that makes the six lines of the model coincide with those of the spectrum. The result of the fit is not a B_{hf} (that is known), but the $Vmax$ that defines the velocity axis for all subsequent spectra.

1. **File > Load...** and select the α-Fe spectrum (for example `data_sample/hierro_metalico_alphaFe.adt`).

2. **Fix the standard and fit the velocity.** Set a single sextet with $B_{hf} = 33{,}0$ T and **mark it as fixed** (padlock), enable *Fit Vmax* and run **Fit > Fit**. The optimizer will move the *Vmax* (and refine δ, Γ) until the six lines match. That *Vmax* value is the calibration.
3. Mark the result as calibration with **File > Use current file as calibration...**, or from the context menu of the file box (right click).

Note

If instead of fixing the field you leave B_{hf} free, you will obtain a value very close to 33 T (e.g. 33,04), but that is redundant: the standard *defines* 33,0 T. Fixing it avoids the degeneracy between B_{hf} and *Vmax* (both scale the line positions) and makes the fit determine *only* the velocity.

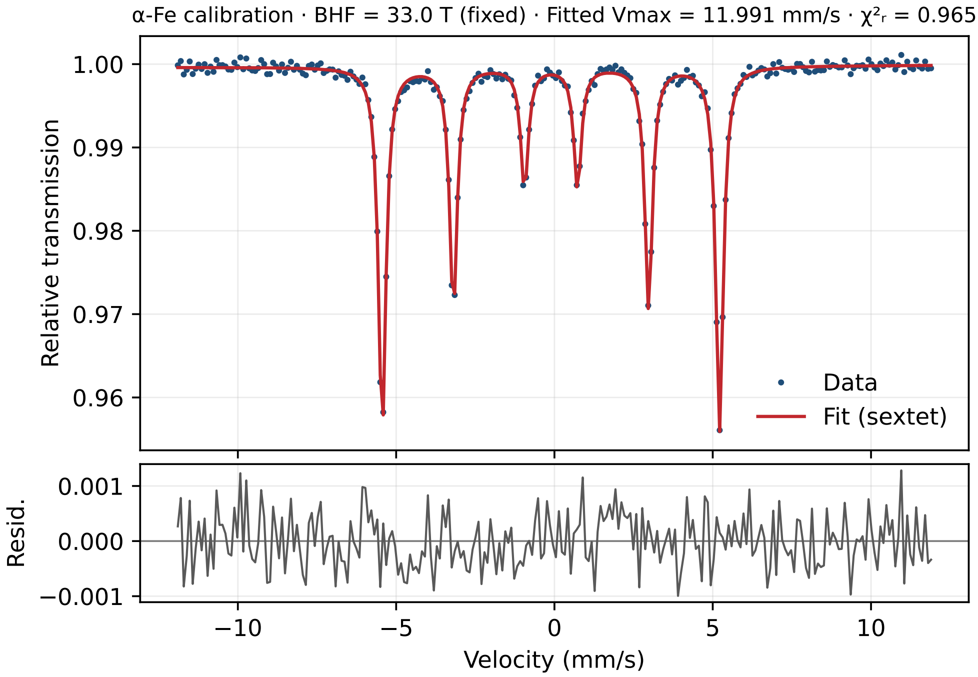

Figure 2.1: Calibration of an α-Fe: $B_{hf} = 33{,}0$ T (standard) is **fixed** and the *Vmax* is **fitted**. The fit returns Vmax $\approx$ 11,99 mm/s with a reduced χ^2 close to 1: that *Vmax* is the calibrated velocity scale. Figure generated with `scripts/make_figures.py`.

When marking the calibration, Fitbauer offers two variants from the context menu of the file box:

- **Use as calibration (current vmax and iso)**: takes directly the *Vmax* and the isomer shift δ of the current state.
- **Use as calibration (details)...**: opens a dialog to review and edit the sample name, the *Vmax* and the IS before fixing.

2.3 The calibration remains *fixed*

Once defined, the calibration remains **fixed**: its *Vmax* (and the reference IS) are saved and shown in the calibration label with the **"fixed"** mark. From that point on it is applied automatically to the spectra you load, according to this rule:

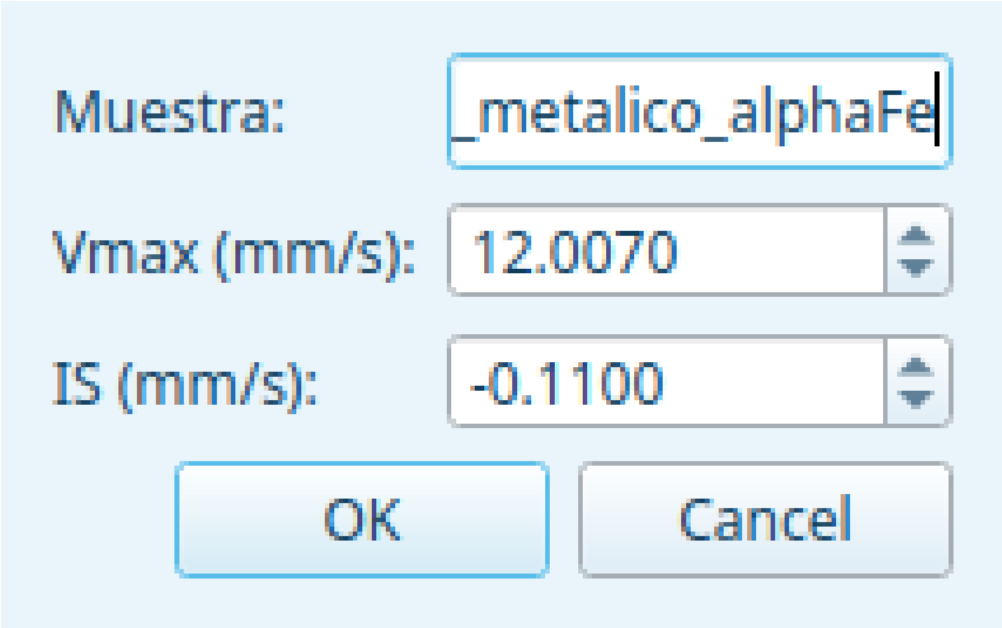

Figure 2.2: "Use as calibration" dialog with the sample name, the calibrated Vmax and the reference isomer shift.

Action	Effect on the calibration
Load a file **without** its own calibration (raw counts: `.ws5`, `.adt`)	The **fixed one is used**: the velocity axis is built with the fixed *Vmax*. The calibration **does not change**.
Load a file **with** its own calibration (velocity axis: `.csv`, `.txt`, `.dat`, `.exp`)	It **replaces** the calibration with the file's own (its *Vmax*) and a **warning** is shown in the status bar.

That is: if you fix a calibration and then load **only data** of raw counts, the calibration remains **intact** and is applied to all of them.

Note

Raw-count files (`.ws5`/`.adt`) do not carry a velocity axis; they need the calibration's *Vmax* to build it. Files already folded in velocity (`.csv`, etc.) bring their own axis: that is why they *replace* the active calibration.

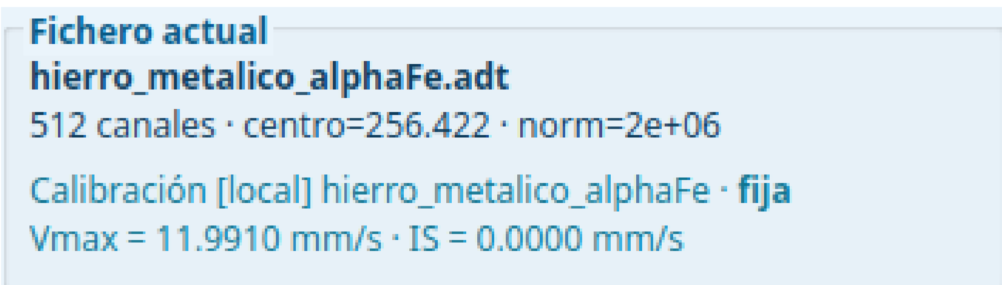

Figure 2.3: Calibration label showing the "fixed" state, with the sample, the Vmax and the reference IS.

2.3.1 Replacing the calibration

When you load a file with its own calibration while another one is already fixed, the new one replaces the previous one and Fitbauer shows a temporary warning in the status bar of the type:

```
Calibration replaced by the file's datos.csv:  Vmax 12.0070 → 11.5000 mm/s
```

This is a record that the active *Vmax* has changed. The reference isomer shift is preserved.

2.3.2 Releasing the calibration

To stop applying the fixed calibration, use **Release fixed calibration** in the context menu of the file box. From then on each load will use the *Vmax* corresponding to its own file (or the control's default value).

2.4 Calibrations from the web

If you have access to the API, **File > Web > Web calibrations...** lets you download a remote calibration. Its values (*velocity_calibrated* and *isomer_shift*) are fixed exactly like a local calibration, rescaling the velocity axis of the current spectrum.

2.5 Persistence in the session

The fixed calibration is part of the **session**: when saving (**File > Save session...**) it is stored together with the rest of the state, and when loading it (**File > Load session...**) it is restored as is. In this way an analysis is reproducible without recalibrating.

Tip

For batch work over many samples measured with the same configuration: load and fix the α-Fe calibration once, save it in a base session and start from it each day.

Chapter 3

Loading and visualising data

With the calibration fixed (chapter 2), loading a spectrum is immediate: Fitbauer folds it, builds the velocity axis and displays it normalised. This chapter describes the formats, the *folding*, the normalisation and the visualisation options.

3.1 Supported formats

Extension	Content
`.ws5`, `.adt`	**Raw counts** per channel (typ. 512 channels). They carry no velocity axis: they are folded and calibrated with the fixed *Vmax*.
`.csv`, `.txt`, `.dat`, `.exp`	Spectrum **already folded** in velocity space (velocity/counts columns). They carry their own axis: they are not folded and *override* the calibration (§2.3).

Note

The `.ws5`/`.adt` files are lists of integers (one count per channel). The velocity files (`.csv`/`.txt`/...) are two columns —velocity and counts/transmission— separated by spaces, tabs or commas; Fitbauer tolerates headers and comment lines. If a velocity file fails to load, check that it has exactly two numeric columns per line.

3.2 Loading a spectrum

File > Load... opens the file selector. When you choose a raw-counts spectrum, Fitbauer:

1. Reads the counts per channel.
2. Detects the **folding centre** (or reads it from a NORMOS `.RES` file if one sits next to the spectrum).
3. **Folds** the two branches of the triangular spectrum and **trims** a few channels from each end (§3.3 explains why).
4. **Normalises** to a baseline of ≈ 1 and estimates the Poisson noise.
5. Builds the velocity axis with the *Vmax* of the fixed calibration.

Note

The status bar and the file label report the number of channels, the detected folding centre and the normalisation factor. **File > Open recent** reopens the most recently used spectra.

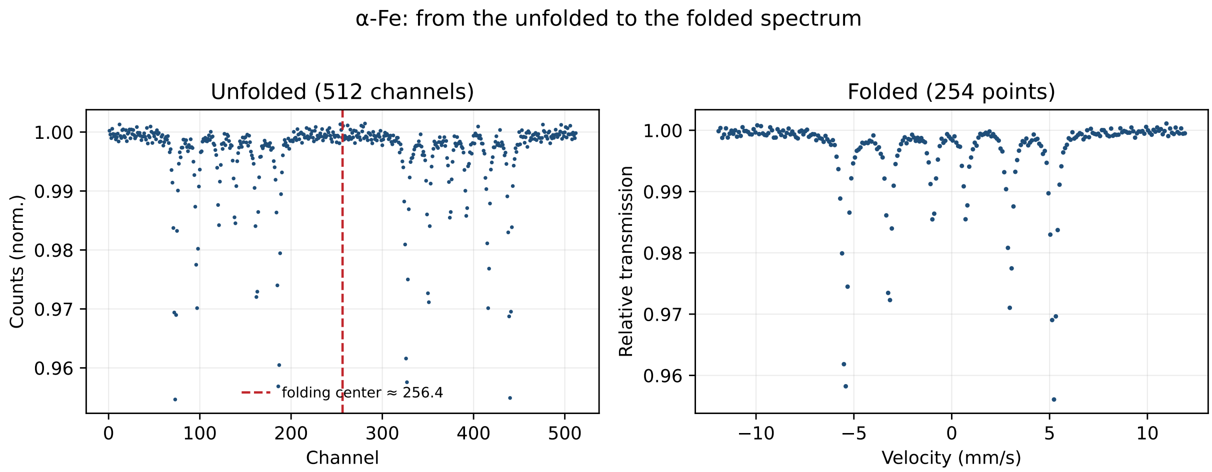

Figure 3.1: The same α-Fe **unfolded** (left, 512 channels) and **folded** (right, in velocity). Unfolded, the transducer measures the sextet *twice* —once in each branch of the sweep— mirrored about the **folding centre** (red line). Folding sums those two halves: it reduces the noise and produces the symmetric spectrum on the right.

3.3 Folding and the centre

The Mössbauer transducer runs through a **triangular** velocity cycle: each velocity is measured twice, once on the rising branch and once on the falling one. ***Folding* sums those two halves** to average them: it improves the signal-to-noise ratio and produces a single symmetric spectrum. The key point is the **centre**: the (possibly fractional) channel at which it is folded.

- Fitbauer detects the centre automatically (or reads it from a `.RES`).
- The *Centre* control of the calibration panel adjusts it by hand: the spectrum is **re-folded live** when it is changed.
- **Fit > Find centre** refines it by minimisation.

Why the ends are trimmed. After folding, Fitbauer discards a few channels from each edge. The reason: the extreme channels correspond to the **turning points** of the transducer (maximum velocity), where the motion reverses direction. There the drive is less linear and there is more uncertainty in the channel→velocity correspondence and in the pairing of the two branches, so those points are the **least reliable**. Trimming them prevents them from contaminating the fit; in exchange a tiny portion of the axis is lost, with no lines of interest. The velocity axis is trimmed at the *same* positions as the data so that both remain aligned (stretching the scale would bias the B_{hf}).

Important

A poorly chosen centre produces double or asymmetric lines. If you observe spurious splittings or a shifted "ghost" spectrum, check the folding centre **first** before touching the model parameters.

3.3.1 Waveform: triangular or sinusoidal

Everything above assumes a **constant-acceleration** drive, whose velocity waveform is **triangular**: the velocity rises and falls in straight ramps, so that it advances *linearly* with the channel and the two halves of the sweep cover the same velocities. That is why it makes sense to fold and use a linear axis.

Some spectrometers use a **sinusoidal** drive instead. Then the velocity of each channel is not linear, but

$$v_i \;=\; v_{\max}\,\sin\!\left(\frac{2\pi\,(i-c_0)}{N}\right),$$

where N is the number of channels and c_0 the **phase** (the channel at which $v = 0$). Since the axis is sinusoidal, folding by symmetry *no* longer gives a linear axis, and forcing a linear one would shift the lines.

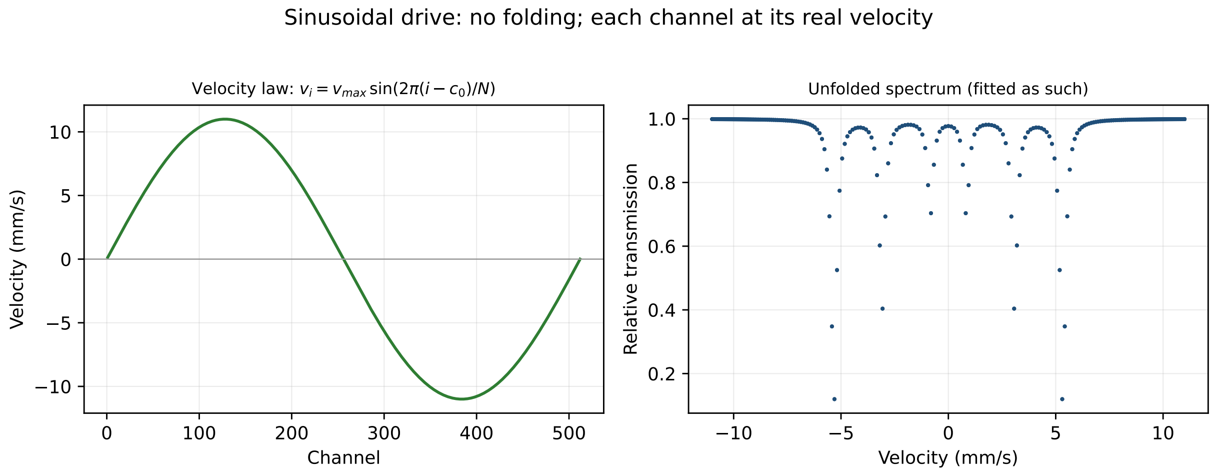

Figure 3.2: Sinusoidal drive. Left: the velocity of each channel follows a *sine*, not a ramp. Right: the resulting spectrum, with each channel placed at its real velocity. It is not folded: it is fitted as is.

The **Waveform** control of the calibration panel selects the mode:

- **Triangular** (default): it is folded and the axis is linear, as described above.
- **Sinusoidal**: following the NORMOS convention (*do not fold* and fit the complete spectrum, with both branches at once), Fitbauer assigns each channel its real sinusoidal velocity and fits **without folding**.

Regarding the **phase** c_0: just as in the triangular case one has to locate the centre, in the sinusoidal case one has to locate the symmetry point of the sweep. Fitbauer autodetects it and exposes it in the *Centre* control (adjustable live). Since a wrongly set phase would shift all the velocities, it is also advisable to **refine it in the fit**: when *Fit centre* is enabled, c_0 enters as a **free parameter** and the optimiser tunes it (just as *Fit Vmax* tunes the amplitude). This is what in NORMOS corresponds to `TRIANG=.FALSE.` with `FOLD=.FALSE.` and `SIMULT=.TRUE.`

Note

In sinusoidal mode the velocity axis is not monotonic (the same velocity appears at several phases of the cycle), so the data are drawn as **points** that run from $-v_{\max}\ldots+v_{\max}$ and back. The model and the fit handle it without problems, because they evaluate the model *point by point* (they do not need an ordered axis). The relaxation component is not supported with a sinusoidal drive in this version.

3.4 Normalisation and noise

So that fits are comparable between spectra, the data are **normalised** by dividing by a factor (the 90th percentile of the folded counts), so that the baseline sits at ≈ 1 and the absorptions read as fractions. That same factor is used to estimate the uncertainty of each point from the Poisson noise (proportional to $\sqrt{N}$), which then weights the fit (see the *likelihood*, §4.4).

Note

The **baseline** and the **slope** (*Slope* control) of the calibration panel model a possible drift of the background. They can be fixed or left free in the fit like any other parameter.

3.5 Visualisation

The plot area shows:

- The **data** (points), the **baseline**, the total **model** and, where appropriate, the **subspectra** of each component (with their optional area fill).
- Below, the **difference plot** (residual) data−model.
- In distribution mode, additionally the **envelope** of the distribution and each sharp component separately, and a panel with $P(B_{\mathrm{hf}})$.

The presentation is controlled from the **View** menu:

- **View > Show difference** / **Show legend** / **Show component area**: enable or hide each element.
- **View > Plot style**: *Classic*, *Modern*, *Publication* or *Dark*.
- **View > Visual theme** and **Colour theme**: general appearance of the interface.
- **View > Configure panel layout...**: rearranges and saves layout presets.

The **toolbar** below the plot (Matplotlib) lets you zoom, pan, return to the initial view and **save the figure** as an image (PNG, PDF, SVG).

3.6 Comparing spectra

File > Compare spectrum... overlays one or several reference spectra (with their own colour palette) on top of the current one, useful for identifying phases or comparing thermal treatments. **File > Clear comparison** removes all overlays. The comparison is purely visual: it does not alter the data or the fit.

3.7 Saving, exporting and reusing

Action	What it saves
File > Save session...	The **complete** state in `.json` (data, fixed calibration, model, options, distribution). Reopen it with **Load session...**.
File > Save fit...	The current fit (parameters, curve, distribution) in text.
File > Export Markdown/PDF report...	Readable report with parameters, statistics and, where appropriate, the distribution table.
File > Export reduced report...	Abbreviated version of the report.

Tip

The **session** is the recommended route for reproducibility: save it when you finish an analysis and you will be able to resume it exactly as it was —including the fixed calibration— at another time or on another computer. If you have the laboratory API, **File > Web** also lets you download spectra/calibrations and **Upload session JSON to web**.

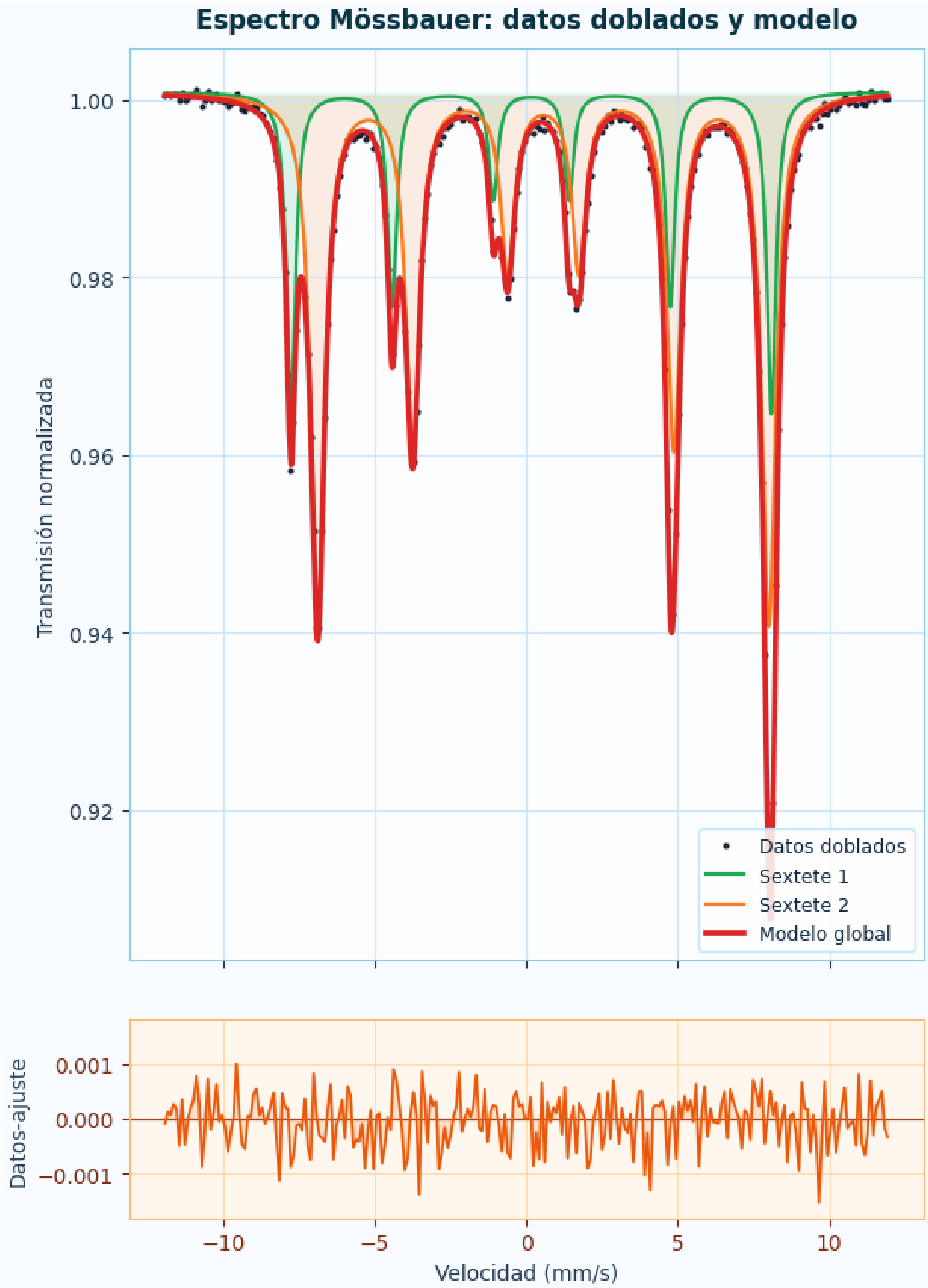

Figure 3.3: Plot area with data, model, subspectra and residual, with its navigation toolbar.

Chapter 4

Simulating and fitting

This chapter covers the *forward model* (simulation) and the *discrete fit*, illustrated with the simplest yet most important case: an α-Fe calibration. Recall the distinction from chapter 1: **simulating** produces a spectrum from parameters that you set; **fitting** infers those parameters from the measured spectrum.

4.1 The forward model

Fitbauer builds the spectrum as transmission:

$$T(v) \;=\; b + s\,v \;-\; \sum_k A_k(v),$$

where b is the baseline, s an optional slope and $A_k(v)$ the *absorption* of each component. The three basic discrete shapes are:

Component	Physical parameters
Singlet	δ (isomer shift), Γ (linewidth), depth. A single peak.
Doublet	δ, ΔE_Q (quadrupole splitting), Γ, depth. Two lines.
Sextet	δ, ΔE_Q, B_{hf} (hyperfine field), linewidths $\Gamma_{1,2,3}$, depth and relative intensities. Six lines.

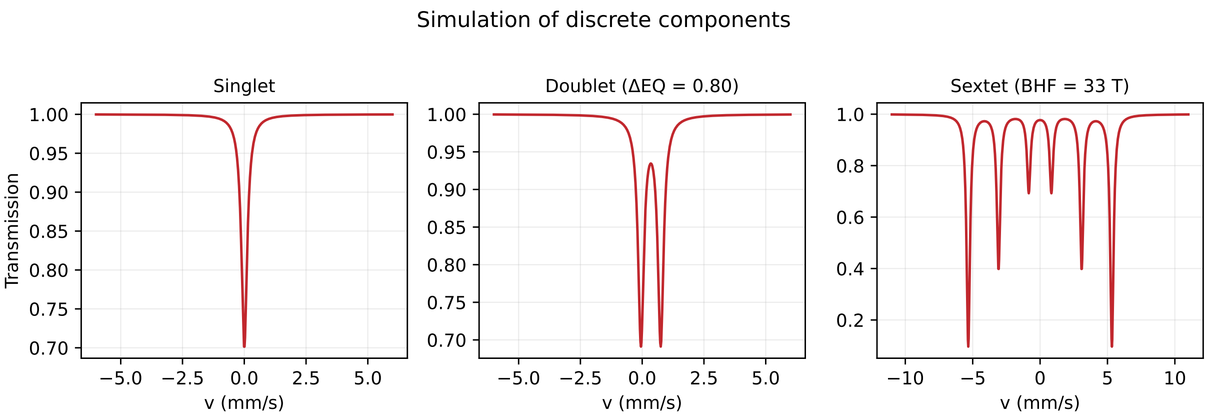

Figure 4.1: Simulation of the three discrete components: singlet (one peak), doublet (ΔE_Q = 0,80 mm/s) and sextet ($B_{\text{hf}} = 33$ T with intensity ratio $3:2:1$).

4.1.1 The mathematics of the model

Each component is a **sum of lines** of Lorentzian shape. A line centred at v_0 with linewidth Γ (the *full width at half maximum*, FWHM —the parameter being fitted—) is

$$L(v;\, v_0, \Gamma) \;=\; \frac{(\Gamma/2)^2}{(v - v_0)^2 + (\Gamma/2)^2},$$

(maximum 1 at $v = v_0$; the half-width is $\Gamma/2$), and the absorption of a component is

$$A_k(v) \;=\; d_k \sum_j w_j \, L(v;\, v_{0,j}, \Gamma_j),$$

with d_k the *depth* and w_j the relative *intensities* of its lines. What distinguishes each type is **where the lines fall** and **how many there are**:

- **Singlet**: one line at $v_0 = \delta$. The isomer shift δ moves the whole pattern.
- **Doublet**: two lines at $v_0 = \delta \pm \frac{1}{2}\Delta E_Q$. The quadrupole ΔE_Q is the separation between the two.
- **Sextet**: six lines
$$v_{0,j} \;=\; \delta \;+\; \tfrac{1}{2}\, q_j \,\Delta E_Q \;+\; p_j \,\frac{B_{\rm hf}}{33{,}0},$$
where $p_j = (\pm 0{,}839,\ \pm 3{,}084,\ \pm 5{,}329)$ mm/s are the published α-Fe line positions at 33 T (which is why the pattern *scales* with $B_{\rm hf}/33$) and $q_j = (+1, -1, -1, -1, -1, +1)$ the first-order quadrupole pattern. The intensities follow the $3 : x : 1 : 1 : x : 3$ ratio (§4.1.3).

The complete spectrum is the transmission $T(v) = b + s\,v - \sum_k A_k(v)$. This parametrisation —positions tied to δ, ΔE_Q, $B_{\rm hf}$ with physical meaning— is what lets you *read* the material directly from the fitted parameters. The full derivation (treatment of the magnetic+quadrupolar Hamiltonian, higher orders) is in appendix A.

4.1.2 Line profile and absorber model

In the calibration panel you choose the **Line profile**:

- **Lorentzian**: the natural shape of the Mössbauer line.
- **Voigt**: convolution of a Lorentzian with a Gaussian of width σ, useful when there is instrumental broadening or a small unresolved spread.

The **Absorber** models the thickness of the sample. The difference lies in how the absorption $A(v)$ is converted into transmission:

- **Thin (linear)**: $T \approx 1 - A(v)$. The absorption is proportional to the amount of Fe; valid for low-density samples.
- **Thick (saturation)**: $T \propto e^{-A(v)}$ (transmission-integral type form). As the thickness increases, the lines **saturate** —they stop growing linearly and broaden—; this mode corrects for that so as not to bias areas or linewidths. The full correction is in appendix F.

Tip

For dense samples or very deep lines ($>$ 15–20%), use **Thick**: with the thin model the relative areas (and therefore the phase proportions) would come out biased.

4.1.3 Sextet intensities and texture

The six lines of the sextet follow the $3 : x : 1 : 1 : x : 3$ area ratio. The value x (lines 2 and 5) depends on the orientation of the moments with respect to the beam:

- In **free** mode, the relative intensities are fitted.
- In **texture** mode, a single parameter t fixes x physically (ratio R_{23}, angle θ), useful for oriented samples or thin films. The panel shows θ and R_{23} derived from t.

The treatment of the **quadrupole** in the sextet can be first-order (perturbative) or complete; it is selected in **Fit > Fit mode**. The detail (Kündig-type treatment of the magnetic+quadrupolar Hamiltonian) is in appendix A.

4.2 Simulating without fitting

Live simulation is one of Fitbauer's hallmarks: instead of launching a fit blindly, you build the model by hand and watch the spectrum change **instantly** with each parameter. To explore the model:

1. Choose the shape of each component in its panel (**Shape:**) and enable it.
2. Set the parameters with the sliders or by typing the value; each control has a *slider*-type bar synchronised with the numeric box.
3. The plot updates **live** with the resulting model and its subspectra.

What simulation is for. Simulation is not just decoration; it serves several practical functions:

- **Seeding the fit**: a non-linear fit needs a reasonable starting point or it falls into a false minimum. Getting the model close "by eye" before **Fit** makes the fit faster and more reliable.
- **Understanding the effect of each parameter**: moving δ shifts the whole pattern; ΔE_Q separates the lines in pairs; B_{hf} opens or closes the sextet; Γ broadens; the depth deepens. Seeing it live builds physical intuition.
- **Anticipating an experiment or comparing hypotheses**: how would a mixture of two phases in these proportions look? You simulate it and compare it with a real spectrum overlaid (§3.6 of the previous chapter).
- **Teaching**: illustrating where each line comes from without needing data.

Each component is drawn as its own **subspectrum** (with its optional area fill), so you can see each one's contribution to the total. The **lock** on each parameter *fixes* it: useful for simulating while varying only one quantity, or so that the subsequent fit does not move what you already know.

Note

Simulating and fitting share exactly the same forward model: what you see when you simulate is, line by line, what the optimiser will try to match. That is why a good initial simulation is the best investment before fitting.

4.3 Fitting an α-Fe calibration

α-Fe is the reference case. The procedure:

1. Load the spectrum (**File > Load...**).
2. Enable a single component of **Sextet** shape.
3. Give approximate initial values: $\delta \approx 0$, $\Delta E_Q \approx 0$, $B_{\mathrm{hf}} \approx 33$ T, $\Gamma \approx 0{,}30$ mm/s.
4. **Fit > Fit**.

The result should reproduce the six lines with $B_{\mathrm{hf}} \approx 33$ T and a reduced χ^2 close to 1, as in figure 2.1 (chapter 2). That fit is, precisely, the one marked as calibration (§2.2).

Tip

If you want the fit to also refine *Vmax* itself against the α-Fe positions, enable *Fit Vmax*. To avoid degeneracy, Fitbauer automatically fixes the active B_{hf} when the velocity is fitted (both cannot be refined at once).

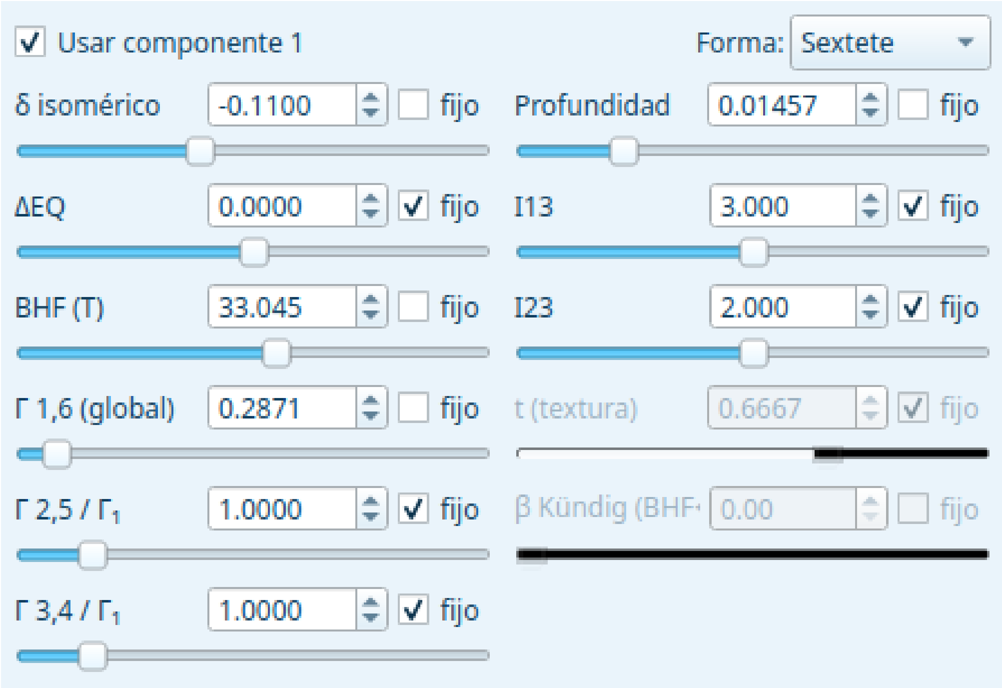

Figure 4.2: Panel of a sextet component with its controls for δ, ΔE_Q, B_{hf}, linewidths and intensities, and the fixing locks.

4.4 Fit options

The statistical criteria are chosen in the **Fit** menu / options panel:

- **Likelihood**: **Gauss** (χ^2 with the data σ) or **Poisson** (IRLS, with the model σ). Poisson is more appropriate with low counts.
- **Robust loss**: **Linear** (least squares), **Soft L1** (smooth) or **Huber**, which downweight outliers.
- **Multistart**: repeats the fit from several starting points so as not to get trapped in a local minimum.
- **Propagate calibration σ (vmax)**: propagates the uncertainty of the calibrated *Vmax* to the fitted parameters.

Note

When *Fit Vmax* is enabled, the calibration takes part in the fit; that is why it is worth being clear about which calibration is set (chapter 2).

4.4.1 Constraints and physical presets

Fit > Constraints between parameters... lets you **tie** parameters (for example, equalising linewidths between components or imposing intensity relations). **Fit > Physical constraint presets...** applies common sets of constraints at a glance. Reducing the number of free parameters stabilises the fit and avoids degeneracies.

4.5 Fit quality and errors

A fit is not finished when the optimiser converges: you have to **judge whether it is good** and **quantify the uncertainty**. For this, Fitbauer gathers several indicators in the information panel and several error methods.

4.5.1 The statistics and what they are for

Indicator	What it measures / what it is for
χ^2	Sum of squared residuals weighted by the noise. It decreases as the fit improves, but it depends on the number of points.
Reduced χ^2	χ^2 per degree of freedom. It is the **main judge**: ≈ 1 = fit compatible with the noise; $\gg 1$ = the model does not reproduce the data (missing components, wrong centre, poor calibration); $\ll 1$ = overfitting or overestimated noise.
dof (degrees of freedom)	Number of points minus free parameters. Context for the χ^2.
AIC and **BIC**	Information criteria: they reward a good fit and **penalise adding parameters**. They serve to **compare models** (e.g. two sites or three?): *lower is better*. They avoid the self-deception that "more components always fit better".
Correlations	Fitbauer warns of pairs of highly correlated parameters (high $\lvert r\rvert$): a sign of **degeneracy** (the fit cannot separate them) or that there are too many parameters. The solution is usually to fix or tie one (§4.4).

Tip

Do not chase a χ_r^2 of exactly 1 by adding components: watch AIC/BIC and the correlations. A model with $\chi_r^2 = 1{,}1$ and well-defined parameters is better than one with $\chi_r^2 = 1{,}0$ full of correlated components.

4.5.2 Estimating uncertainties

The statistics tell you whether the fit is good; the **errors** tell you how much you can trust each parameter. Fitbauer offers three methods, from lower to higher cost and reliability:

Method	What it does / when to use it
Covariance (default)	Errors from the covariance matrix of the fit. Instantaneous; valid if the uncertainties are approximately Gaussian and the correlations moderate. It is the first estimate.
Bootstrap errors (MC)...	Refits many replicas of the spectrum resampled according to the noise (Monte Carlo) and measures the spread of each parameter. More robust against noise and non-linearity; it costs more time.
Profile-likelihood errors...	Scans each parameter, refitting the others, and sees how much the χ^2 worsens (*profile likelihood*). It gives realistic **asymmetric** intervals; the most reliable when the uncertainties are **not** Gaussian or there are strong correlations.

Note

Rule of thumb: use **covariance** during exploratory work and reserve **bootstrap** or **profile likelihood** for the figures you are going to publish, especially if the panel warns of high correlations.

4.6 Fitting-aid tools

- **Fit > Initialise from minima** and **Auto-fit from minima**: they detect the line positions and propose a starting point. The **minima editor** also lets you *curate* them on the plot itself: click to add a minimum, click on a marker to activate/deactivate it, and a counter of contributions per minimum.
- **Fit > Find centre**: refines the folding centre.
- **Fit > Fix all / Release all**: lock or release all parameters at once.
- **Fit > Batch fit...**: fits a batch of spectra by *warm-start* (each fit starts from the previous one), ideal for temperature or treatment series.
- **Fit > Undo fit**: reverts to the previous state.
- **Fit > Local Ollama AI: suggest start...**: if you have a local model, it proposes initial values from a summary of the spectrum.

4.7 Magnetic relaxation components

In addition to singlet/doublet/sextet, Fitbauer includes components for **superparamagnetic relaxation** and spin dynamics, useful in nanoparticles near the blocking temperature:

- **Relaxation** and **Blume–Tjon**: line models with a relaxation rate ν (parameter $\log_{10}\nu$).
- **Néel (size)**: lognormal size distribution with anisotropy K_{eff} and blocking temperature; **Fit > Néel global fit (multi-T)...** fits several temperatures at once.

Note

These models share the component interface (they are chosen in **Shape:**). Their numerical basis is developed in appendix E (*Magnetic relaxation models*).

Figure 4.3: Semi-manual minima editor on the plot: the included (red) and excluded (hollow) markers are edited with the mouse and with the side list.

Chapter 5

Hyperfine field distributions

If you have made it this far, you know how to recognize and fit a singlet, a doublet or a sextet: you know that each one corresponds to a **well-defined iron site**, with a single isomer shift, a single quadrupole and—in the sextet—a single hyperfine field. *Distributions* are the next step, and their logic is different. This chapter starts **with the physics**: what they are, where they come from and what each parameter means. Only afterwards does it address the program's controls.

5.1 The physics: why distributions appear

5.1.1 From the perfect crystal to the real material

In a **perfect crystal** all the Fe atoms are equivalent: they see the same environment, so they have *exactly* the same hyperfine field B_{hf}, the same δ and the same ΔE_Q. The result is a **discrete**, sharp sextet (or doublet, or singlet) of narrow lines. This is the world of the previous chapters.

But most materials **are not perfect crystals**. In a metallic glass, a disordered alloy, a non-stoichiometric oxide or a nanoparticle, each Fe atom sees an environment **slightly different** from that of its neighbor: it may have a different number of atoms of a certain type around it, a different bond distance, a different local strain. And since the hyperfine parameters *depend* on that environment, they no longer have a single value: they **spread out** over a range.

> **Note**
>
> The key idea: in a disordered material there is no single hyperfine field, but rather **as many fields as there are distinct local environments**. The spectrum ceases to be a sextet and becomes the **superposition of very many sextets**, each with its own field, weighted by how many Fe atoms have that field.

5.1.2 What $P(B_{\mathrm{hf}})$ is

That "how many atoms have each field" is precisely the distribution. $P(B_{\mathrm{hf}})$ is a **probability density**: $P(B_{\mathrm{hf}})\,dB_{\mathrm{hf}}$ is the fraction of Fe atoms whose hyperfine field falls between B_{hf} and $B_{\mathrm{hf}} + dB_{\mathrm{hf}}$. Put another way, it is the **histogram of the local environments** seen by the iron.

- A **crystalline** material has a *very narrow* $P(B_{\mathrm{hf}})$ (almost a delta): nearly all atoms share the same field.
- A **disordered** material has a *broad* $P(B_{\mathrm{hf}})$: the fields spread out over a wide interval.
- A material with **two populations** (e.g. two phases, or the core and surface of a nanoparticle) has a $P(B_{\mathrm{hf}})$ with **two maxima**.

The measured spectrum is then

$$T(v) \;=\; b \;-\; \int P(B_{\mathrm{hf}})\; S(v; B_{\mathrm{hf}})\, dB_{\mathrm{hf}},$$

the sum (integral) of all the subspectra $S(v; B_{\rm hf})$ weighted by $P(B_{\rm hf})$. Fitting a distribution is **the inverse problem**: starting from the spectrum and recovering the $P(B_{\rm hf})$ that generated it.

Note

That is why a distribution is *not simulated* out of nothing: it is **inferred from the spectrum** (fitting). What you can simulate is the *spectrum* that a $P(B_{\rm hf})$ you *assume* would produce—useful for developing intuition (§5.3). The mathematical development of the inverse problem is in appendix A.

5.1.3 A distribution is not the same as a broad line

This point is often confused and it is worth pinning down. There are **two ways** for the lines of a spectrum to come out broad, and they mean different physical things:

- **Line width** Γ: the *intrinsic* broadening of *one* environment (lifetime of the state, instrumental effects). It broadens **all** the lines equally and symmetrically, without changing their positions.
- **Distribution width**: the *range of distinct environments*. Since the position of the sextet lines scales with the field (§4.1.1), a spread of fields broadens **the outer lines more than the inner ones**, and can produce **asymmetric** shapes.

Important

Do not try to "cover up" a distribution by increasing Γ. A single sextet with large Γ and a genuine broad $P(B_{\rm hf})$ do **not** give the same spectrum: the distribution broadens the outer lines more than the inner ones and can be asymmetric, something Γ does not reproduce. If, when fitting discretely, the widths come out "inflated" and the outer lines fit worse than the inner ones, it is a sign that there is a distribution.

5.1.4 What each feature of the distribution tells you

Once $P(B_{\rm hf})$ is obtained, its **shape** is physical information:

Feature of $P(B_{\rm hf})$	**What it implies physically**
Position (mean / maximum)	The *typical* field: reflects the average environment (composition, coordination, average magnetic state).
Width	The **degree of disorder**: the broader it is, the more the environments vary (more chemical or structural spread).
Asymmetry (tail)	A gradient of environments: e.g. a minority of atoms with many neighbors of another type generates a tail toward high or low fields.
Several maxima	Distinct **subpopulations**: two phases, two crystallographic sites, nanoparticle core/surface, etc.
Weight at $B_{\rm hf} \approx 0$	**Non-magnetic / relaxing** fraction (paramagnetic or superparamagnetic at that temperature).

The same reasoning applies to $P(\Delta E_Q)$ (spread of electric field gradients in paramagnetic materials) and $P(\mathrm{IS})$ (spread of electron densities / oxidation states), covered in appendix C.

5.2 Idea: from a sextet to a distribution

A distribution shares out the probability among a grid of fields $\{B_{\rm hf}^{(j)}\}$ with weights $P_j \geq 0$. The spectrum is

$$T(v) = b - \sum_j P_j S_j(v),$$

where S_j is the sextet corresponding to the field $B_{\rm hf}^{(j)}$. A **crystalline material** with a single field is represented as a *narrow* distribution (almost a delta) about that value; a disordered material, as a *broad* distribution.

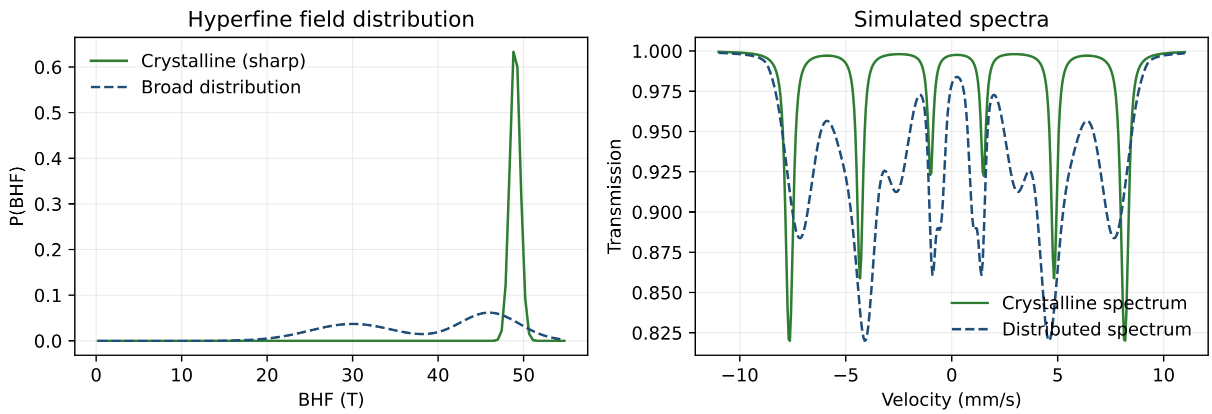

Figure 5.1: Left: crystalline hyperfine field distribution (sharp, around 49 T) versus a broad distribution. Right: the corresponding simulated spectra. The broad distribution “blurs” and overlaps the sextet lines.

Notice in the figure how a narrow $P(B_{\rm hf})$ produces a sextet of narrow lines (crystalline behavior), whereas a broad one gives a spectrum with broadened and overlapping lines.

5.3 Simulating the spectrum of a crystalline case

The crystalline case (a well-defined field) is the natural starting point for understanding the model. In distribution mode:

1. Enable **distribution mode** and choose the variable ($P(B_{\rm hf})$ or $P(\Delta E_Q)$).
2. Set the range of the grid (minimum and maximum $B_{\rm hf}$) and the number of bins.
3. Set the shared kernel parameters: δ, ΔE_Q and the line width Γ.
4. Assume a *narrow* distribution centered on the expected field (almost a delta): the simulated spectrum will tend toward that of a crystalline material, with narrow lines.

The plot shows at the same time the assumed **distribution** $P(B_{\rm hf})$ and the **spectrum** it produces, so that you can see the effect of narrowing or broadening the distribution. On real data, by contrast, $P(B_{\rm hf})$ is **fitted** (following sections).

5.4 The distribution panel, control by control

The distribution panel brings together quite a few controls. Here **each one** is explained: what it is, what it is for, what values are reasonable and when it is worth adjusting. Many are **hidden or disabled** depending on the chosen shape (for example, the regularizer only appears with Histogram), so do not be alarmed if you do not see them all at once.

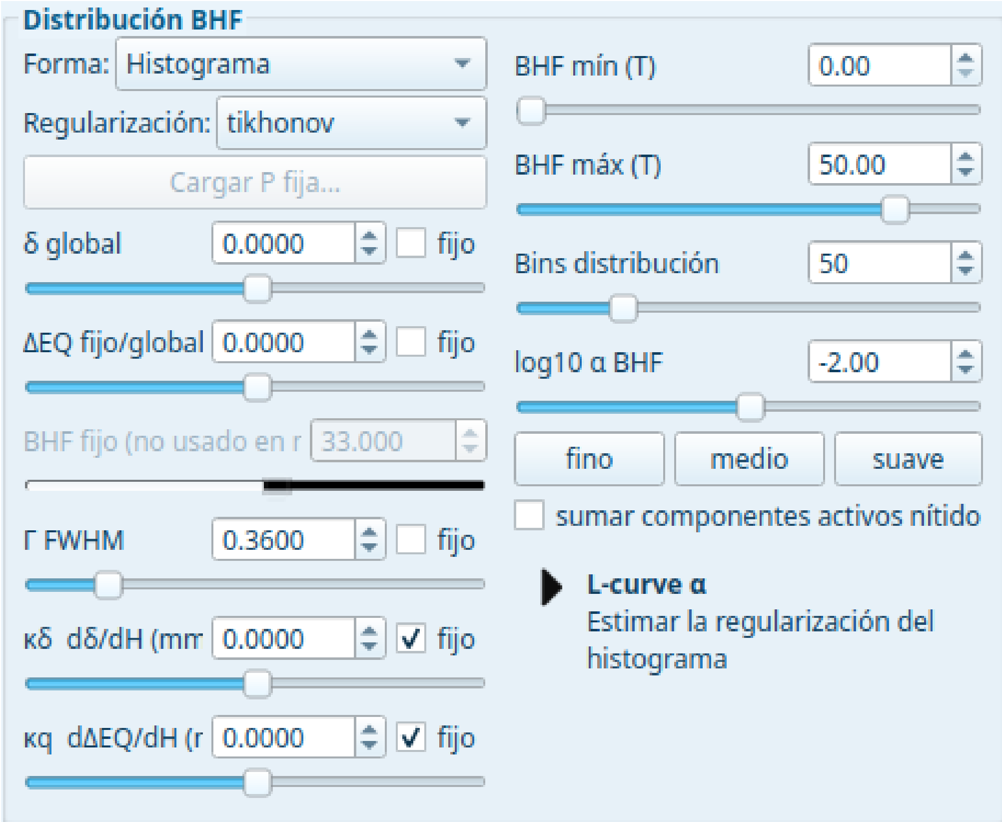

Figure 5.2: Distribution panel: variable, shape, regularizer, grid range and number of bins.

Note

Underlying idea: distribution fitting builds **many subspectra** (sextets or doublets), one for each value of the variable on a grid, and looks for **what weight** $P_j \geq 0$ each one has in order to reproduce the spectrum. The panel parameters control (a) *what* is distributed, (b) *how* each subspectrum looks, (c) the *grid* of values and (d) the *regularization*.

5.4.1 What is distributed: the variable

The first thing is to decide **which quantity** varies from one environment to another. It is chosen in the mode selector:

Mode	What it distributes
$P(B_{\mathrm{hf}})$	The **hyperfine field** (magnetic materials: each environment has its own field). It is the most common case.
$P(\Delta E_Q)$	The **quadrupole splitting** (paramagnetic materials: distribution of electric field gradients, with doublets instead of sextets).
$P(\mathrm{IS})$	The **isomer shift**.
2D	Joint distribution of two variables at once, e.g. $P(B_{\mathrm{hf}}, \Delta E_Q)$ (§5.9).

The distributed variable is the one that sweeps the grid; the *other* quantities are fixed as global kernel parameters (next subsection).

5.4.2 Shape and regularizer

- **Shape**: the family of P that is fitted—Histogram, Gaussian, VBF, Binomial, Fixed or 2D—(§5.5 details them). It is the decision that most conditions the rest of the controls.
- **Regularizer**: only with **Histogram**. Chooses how the lack of smoothness is penalized—*tikhonov*, *tv* or *maxent*— (§5.6).
- **VBF components**: only with **VBF**. Number of Gaussians (1–6) that make up the distribution. Start with 1–2 and add more only if the fit calls for it.

5.4.3 Kernel parameters (each subspectrum)

Each subspectrum of the grid shares these "line" parameters. They are the ones that define *how* an individual sextet/doublet looks before sharing out the weights:

Control	Function	Typical	Range
δ	Isomer shift common to all subspectra (shifts the pattern). Fix it to the δ of your phase or leave it free.	0	$-2{,}5 \ldots 2{,}5$
ΔE_Q	Global quadrupole (in $P(B_{\mathrm{hf}})$ mode). It is **disabled** if it is the distributed variable.	0	$-4 \ldots 4$
Fixed BHF	Fixed field used when ΔE_Q or IS is distributed (not used in $P(B_{\mathrm{hf}})$ mode).	33	$0 \ldots 60$
Γ	**Line width** of each subspectrum. *Key*: it controls the resolution (see the warning).	0,30	$0{,}06 \ldots 2{,}0$

Important

The width Γ and the number of bins are **coupled** with the regularization. If you set Γ too small, each subspectrum is very narrow and the histogram tends toward "spikes" and to follow the noise; if you set it large, the distribution looks smoother but loses detail. A physically reasonable value ($\Gamma \approx 0{,}25$–$0{,}35$ mm/s for ^{57}Fe) and an α chosen with the L-curve give the best compromise.

5.4.4 Correlation with the field: κ_δ and κ_q

Two optional slopes that couple δ and ΔE_Q to the value of the field (§5.8). By default they are **0** (no correlation) and are **fixed**; to refine them, uncheck their lock. They only apply to Histogram and VBF.

5.4.5 The grid of values

Defines over which values of the variable the probability is shared out:

Control	Function	Typical	Range
min	**Lower** end of the grid.	0	$0 \ldots 60$
max	**Upper** end. Set it to the range where you *expect* fields (e.g. 0–55 T).	50	$0 \ldots 60$
bins	**Number of points** of the grid. More bins = more resolution but a more ill-conditioned problem (more regularization needed).	50	$10 \ldots 200$

Tip

Choose the **range** as tight as possible to where there is signal: a grid that extends over empty zones "wastes" bins and can introduce spurious structure at the edges. With 40–80 bins is usually enough; raise it only if you need to separate fine details.

5.4.6 The regularization: $\log\alpha$ and shortcuts

Only for Histogram (and 2D):

- **log**$_{10}\alpha$: strength of the regularization (§5.6). α is dimensionless: $\log\alpha \approx 0$ is the natural balance. Less $\alpha \to$ more detail and more noise; more $\alpha \to$ smoother.
- [Fine] / [Medium] / [Smooth] buttons: shortcuts that set $\log\alpha$ to progressively larger values (from more detail to more smoothness), for quick trials.
- [L-curve α]: opens the dialog that **chooses α with a criterion** (§5.7). It is the recommended route.

5.4.7 2D controls

They appear only with the **2D** shape: **min/max/bins of the second axis** (e.g. ΔE_Q) and **log**$_{10}\alpha$ **of the second axis**, analogous to those of the first variable but for the Y axis. The [View 2D map] button opens the map (§5.9).

5.4.8 Other

- **Add active sharp components**: fits the distribution and the active discrete components at the same time (§5.10).
- **Load fixed P. . .**: only with the **Fixed** shape; reads a P from a file (two columns: value and weight).

5.5 Distribution shapes

Fitbauer offers several **shapes** (**Shape:**) for $P(B_{\mathrm{hf}})$. Some are *non-parametric* (they leave the weight of each bin free) and others *parametric* (they impose a function with few parameters):

Shape	Description
Histogram	Hesse–Rübartsch inversion: free weights per bin with regularization. The most general and flexible; requires choosing α.
Gaussian	A single Gaussian in B_{hf} (mean and width). Rigid but very stable.
VBF	Sum of several Gaussians (Voigt-Based Fitting, Rancourt–Ping); saves A_k, μ_k, σ_k per component (§5.5.7).
Binomial	Binomial distribution (neighbor-environment models, e.g. alloys).
Fixed	Distribution imposed by the user (read from a file).
2D	Joint distribution $P(B_{\mathrm{hf}}, \Delta E_Q)$ (§5.9).

Tip

Choose **Gaussian** or **VBF** when you expect one or a few well-defined environments (they give parameters with direct meaning); reserve the **Histogram** for complex or unknown shapes, where you do not want to impose a function.

Each shape is detailed below: what model it fits, what controls it uses and when to choose it.

5.5.1 Histogram (Hesse–Rübartsch)

This is the **non-parametric** shape and the most general. It imposes no function: it treats the weight of *each bin* as an independent unknown. The spectrum is written as a linear combination of the subspectra of the grid,

$$T(v) \;=\; b + s\,v \;-\; \sum_{j=1}^{N_{\text{bins}}} P_j\, S_j(v), \qquad P_j \geq 0,$$

and fitting means **solving for the weights** $\{P_j\}$ that best reproduce the data. It is a linear problem with non-negativity, but **ill-conditioned** (many $\{P_j\}$ give almost the same spectrum), hence the need to **regularize** (§5.6). The method goes back to Hesse and Rübartsch (1974).

- **Uses**: grid (min/max/bins), kernel (δ, ΔE_Q/fixed BHF, Γ), **regularizer** and $\log\alpha$, correlations κ_δ/κ_q.
- **When**: when the shape of P is *unknown* or complex (several environments, asymmetric tails) and you do not want to impose a function.
- **Price**: you have to choose α well (use the L-curve, §5.7); it is the only shape that needs it.

5.5.2 Gaussian

Minimal **parametric** shape: P is *a single Gaussian* over the grid. It fits its **mean** μ (the central field) and its **width** σ (plus the amplitude and the baseline). Only two shape parameters, so it is **very stable** and does not suffer from ill-conditioning: it uses no regularizer or α. Internally it is exactly the **VBF with a single component** ($N = 1$, §5.5.3 and appendix B): they share the algorithm, and choosing "Gaussian" is equivalent to fixing $N = 1$ with a Lorentzian line.

- **When**: there is *one* broadened environment (a phase with an approximately symmetric spread of fields), or as a quick **first approximation** before complicating the model.
- **Gives**: μ and σ with direct physical meaning.

5.5.3 VBF (Voigt-Based Fitting, Rancourt–Ping)

Generalizes the Gaussian to a **sum of** N **Gaussians** (**VBF components**, $N = 1\ldots6$). Each component k has amplitude A_k, mean μ_k and width σ_k, all adjustable:

$$P(B_{\text{hf}}) \;=\; \sum_{k=1}^{N} A_k\, \mathcal{N}(B_{\text{hf}};\, \mu_k, \sigma_k).$$

This is the method of Rancourt and Ping (1991). Its great advantage over the Histogram: it delivers **parameters with meaning** per subpopulation (position, width and **area**, §5.5.7) instead of a structureless profile, and it is more stable (few parameters, no regularization). Appendix B develops its mathematics (direct model, Voigt profile, parametrization, populations and the case $N = 1$ that corresponds to the **Gaussian** shape).

- **When**: you expect **2–3** well-defined magnetic **subpopulations** and want to quantify their *proportion*.
- **Advice**: start with $N = 1$–2 and raise it only if the residual calls for it; more Gaussians than needed overlap and end up poorly defined.
- **Supports**: correlations κ_δ/κ_q.

5.5.4 Binomial

Parametric shape with a single shape unknown: P follows a **binomial distribution** $B(n, p)$ over the grid (with $n = N_{\text{bins}} - 1$), and the **probability** $p \in (0, 1)$ is fitted (the mean falls at $p\,n$).

Its motivation is physical: in an **alloy**, the hyperfine field of an Fe atom depends on **how many solute atoms** it has in its neighbor shells. If the solute is distributed at random, that number follows a binomial, and therefore so does the distribution of fields. The parameter p is related to the **concentration** of solute.

- **When**: *neighbor* systems (dilute alloys) where each neighbor configuration gives a discrete field.

5.5.5 Fixed

Uses a distribution P **that you provide** in a two-column file (value and weight), loaded with **Load fixed P...**. The shape is *not* fitted (only amplitude and baseline): it serves to **reuse** a known P—from a previous fit, from the literature or theoretical—and compare or simulate with it over different spectra.

5.5.6 2D: two variables at once

Distributes **two** quantities jointly (e.g. $P(B_{\text{hf}}, \Delta E_Q)$) over a two-dimensional grid. It is the most general and most costly case; it is treated separately in §5.9.

5.5.7 Areas of each Gaussian (VBF)

In **VBF** mode with more than one Gaussian, each component contributes an area A_k (its integral over the grid, because the Gaussian kernel is normalized to unit area). Fitbauer reports that **area** and its **relative percentage** $A_k / \sum_j A_j$, which represents the **population of each environment**:

- In the **status panel**: keys `vbf`k`_area`, `vbf`k`_area%` and `vbf`k`_mu` per component.
- In the **fit summary** and in the **reports** (table with A, area %, μ and σ).

This makes VBF a convenient route to quantify the proportion of two or more magnetic subpopulations without resorting to a histogram.

5.6 Histogram regularization

Fitting a **Histogram** to real data is an *ill-conditioned* problem: many different $P(B_{\text{hf}})$ reproduce the spectrum almost equally well, and without control spurious oscillations appear. The solution is to **regularize**: add to the fit a term that penalizes solutions that are not smooth, controlled by a parameter α.

5.6.1 Regularizers

Regularizer	Effect / when to use it
Tikhonov	Penalizes the curvature (second difference): *smooth* solutions. Default option, suitable for broad, rounded distributions.
TV (total variation)	Penalizes the total variation: preserves *edges* and steps; useful if you suspect well-separated populations.
MaxEnt (maximum entropy)	Seeks the most "uniform" distribution compatible with the data (L-BFGS-B with analytical gradient); avoids introducing unjustified structure.

5.6.2 The parameter α

α balances **fidelity to the data** (small $\alpha \to$ follows the noise, rough distribution) against **smoothness** (large $\alpha \to$ smooth distribution but a worse fit). In Fitbauer α is **dimensionless**: $\log_{10}\alpha \approx 0$ is the natural balance for any spectrum, and the useful values usually fall between -1 and $+2$.

Important

α is the most delicate choice in histogram fitting. Do not leave it to chance: use the **L-curve** (§5.7) to choose it with a criterion.

5.7 The L-curve: choosing α with a criterion

Choosing α "by eye" is risky: a value that is too low injects noise into the distribution and one that is too high flattens it and hides real structure. The **L-curve** provides an objective criterion. It is the most important tool in histogram fitting.

5.7.1 What it is and why it has an L shape

For each α in the sweep, the fit produces a distribution P_α, and from it **two** competing quantities are measured:

- **Misfit** (residual) $\|KP_\alpha - y\|$: how much the model departs from the data. We want it to be *small*.
- **Roughness** $\|L\,P_\alpha\|$: how much the distribution "trembles" (the norm of the smoothness operator L). We also want it to be *small*.

Both cannot be minimized at once: lowering α improves the fit but dirties the distribution, and raising it does the opposite. If **roughness against misfit** is plotted on a log-log scale as α is varied, a curve with an **L** shape appears:

- **Vertical branch** (α small, *under-regularized*): continuing to lower α barely improves the fit, but the roughness **shoots up**—the solution starts to fit the noise.
- **Horizontal branch** (α large, *over-regularized*): the distribution is already smooth; raising α further barely smooths it, but the fit **gets worse**—real structure is lost.
- **The corner**: the point where the curve turns. It is the **best compromise**: the smallest possible α that still keeps the distribution reasonably smooth.

Formally, the corner is the point of **maximum curvature** of the L (Hansen's criterion). Fitbauer detects it automatically (Menger curvature over triples of points) and marks it in green as a *suggestion*.

5.7.2 The dialog, panel by panel

- **Left — the L**: roughness against misfit (log-log), with each point labeled by its $\log\alpha$. The automatic corner in green and the **chosen** point in orange.
- **Right — α vs RMS**: the fit error (RMS) against α; it helps to see where the fit starts to degrade.

Both figures are **coupled**: when you zoom on one, the other is limited to the same α range. The toolbar allows you to zoom into the corner region.

5.7.3 Choosing α: automatic or by hand

The automatic corner detection **does not always get it right** (noisy curves, poorly marked corners), so the dialog is **interactive**:

- **Click** on either of the two figures to **choose** the scanned α closest to the click; the point is highlighted in both.

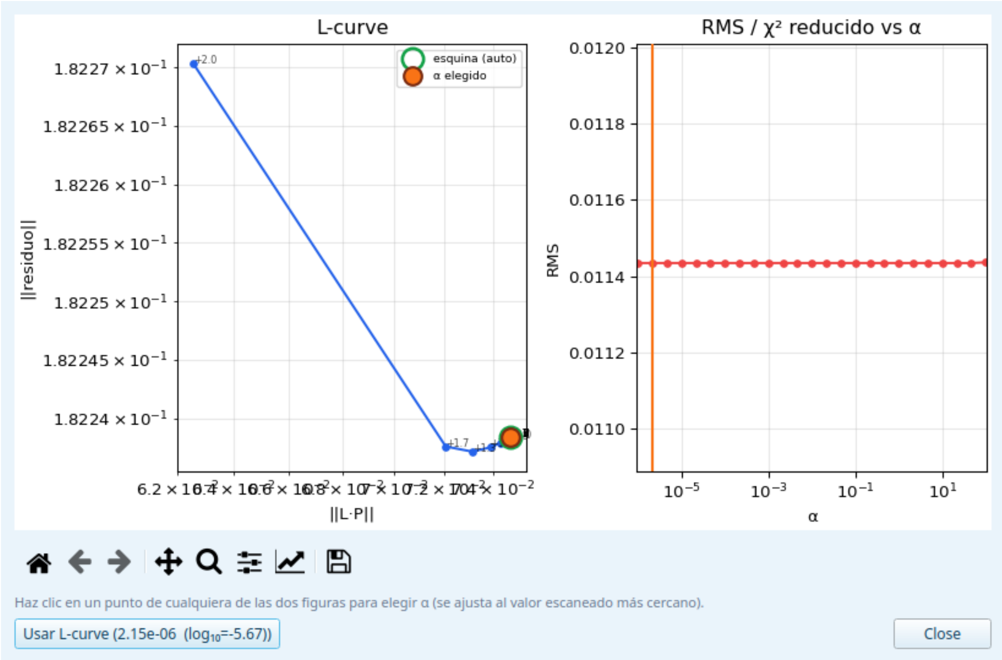

Figure 5.3: L-curve dialog: on the left the roughness↔misfit curve with the automatic corner (green) and the chosen point (orange); on the right α against RMS. A click chooses α.

- The [Use α=. . .] button applies the *chosen* value (it starts at the automatic suggestion) to the fit and closes the dialog.

5.7.4 How to interpret it in practice

1. Accept the **automatic corner** as a first approximation and observe the resulting $P(B_{\mathrm{hf}})$.
2. If you see it **too rough** (spikes that look like noise), raise α a little (click a point to the right on the L). If you see it **too washed out** (it has lost detail that the residual revealed), lower α (a point to the left).
3. Check the **stability**: the correct solution changes *little* when you move α one step up or down. If it changes a lot, you are on a branch, not at the corner.

Important

Sometimes **there is no clear corner** (the L is rounded). This usually means that the data do not demand much structure: choose an α from the transition zone and validate it by the stability of $P(B_{\mathrm{hf}})$ and by the χ_r^2 of the fit, not by the corner. And remember: the L-curve only applies to the **Histogram** (the parametric shapes do not use α).

5.8 Correlation $\delta(H)$ and $\Delta E_Q(H)$

So far we have assumed that δ and ΔE_Q are **the same for all** the subspectra of the distribution, and only the field varied. But physically that is not always true: the disorder that spreads the field usually spreads δ and ΔE_Q *as well*, and in a **correlated** way. For example, an Fe atom with more neighbors of a certain type has at once a different field *and* a different electron density (a different δ): the two quantities move together.

Fitbauer captures that correlation with two **slopes** (*opt-in*, zero = classical behavior without correlation):

- $\kappa_\delta = d\delta/dH$: makes $\delta(H) = \delta_0 + \kappa_\delta\,(H - H_{\mathrm{ref}})$, that is, the isomer shift *varies linearly* with the field of the subspectrum.

- $\kappa_q = d\Delta E_Q/dH$: the same for the quadrupole.

What it implies. With $\kappa_\delta \neq 0$, the high-field subspectra use a different δ than the low-field ones, so that the distribution no longer just "opens" the lines but also *shifts* them in a field-dependent way. It is the way to model, with *one* extra parameter, that the chemical and magnetic environments change together. If your material requires it, you will see the residual improve when κ_δ/κ_q are freed (by unchecking their lock).

They apply to the **Histogram** and **VBF** shapes. The fitted values of κ_δ and κ_q are shown in the status panel and in the reports.

Tip

Always start with $\kappa = 0$. Activate them only if, after a good fit without correlation, the residual shows a systematic pattern that suggests δ or ΔE_Q depend on the field. A misused κ can artificially compensate for other defects of the model.

5.9 2D distributions: $P(B_{\mathrm{hf}}, \Delta E_Q)$

When field and quadrupole vary in a way that is **not correlatable with a straight line** (that is, when κ_δ/κ_q are not enough), the **2D** shape fits a joint density $P(B_{\mathrm{hf}}, \Delta E_Q)$ over a two-dimensional grid. Each axis has its own controls:

- X axis (e.g. B_{hf}): the usual **min/max/bins** and **log**$_{10}\alpha$.
- Y axis (e.g. ΔE_Q): **min/max/bins of the second axis** and its own **log**$_{10}\alpha$, which is **regularized separately**.

The result is shown as a **map** (*heatmap* with contours)—accessible with [View 2D map]—in addition to the marginal distributions of each variable. The mathematical development of the two-dimensional case is in appendix D.

Important

The 2D has **many more unknowns** (one per grid cell) than the 1D, so it is more costly and harder to regularize. Start with **few bins** on each axis (e.g. 15×15) to test, check that the map is stable, and only then refine the grid. Two α values that are too low produce a map full of false peaks.

5.10 Sharp components alongside the distribution

It is common for a sample to combine a **crystalline phase** (a sharp sextet) with a **distributed phase**. With **P(BHF): add active sharp components**, one or several discrete components and the distribution are fitted **simultaneously**. The plot then draws each sharp component separately, the **envelope** of the distribution and the total model, so that the contribution of each phase can be seen.

5.11 Step by step: fitting a $P(B_{\mathrm{hf}})$ from scratch

If this is your first time, follow this recipe. Each step refers to the section that explains it in detail.

1. **Load and calibrate.** Open the spectrum and make sure you have a fixed calibration (chapter 2); without the correct velocity scale, the fields will come out biased.
2. **Enter distribution mode** and choose the **variable** $P(B_{\mathrm{hf}})$ (§5.4.1).
3. **Test with a simple shape.** Start with **Gaussian** (or **VBF** with 1–2 components): they are stable and give you a quick idea of where the field is and how broad it is. Fix δ to that of your phase and a $\Gamma \approx 0{,}30$ mm/s (§5.4.3).

4. **Adjust the grid range** (min/max) so that it covers where there is field, without excess (§5.4); use ~ 50 bins.
5. **Launch the fit** (**Fit > Fit**). Look at $P(B_{\text{hf}})$ and the residual: does it reproduce the lines?
6. **Complex shape?** If one or a few Gaussians are not enough, switch to **Histogram** (§5.5), which imposes no function.
7. **Choose α with the L-curve** (§5.7): do not leave it at the default. Accept the corner or click a point and compare the $P(B_{\text{hf}})$.
8. **Refine the model** if appropriate: activate **sharp components** for a superimposed crystalline phase (§5.10), or the slopes κ_δ/κ_q if $\delta/\Delta E_Q$ depend on the field (§5.8).
9. **Document.** With VBF, note the **areas** of each Gaussian (phase proportion, §5.5.7). Save the **session** and export the **report** with the distribution table.

Tip

Rule of thumb: go **from simple to complex**. A stable Gaussian that explains 95% of the spectrum is a better starting point than a poorly regularized Histogram. Increase in complexity (VBF $\rightarrow$ Histogram $\rightarrow$ 2D) only when the residual justifies it, and always watch that the solution is **stable** against small changes in α, Γ or number of bins.

Appendix A

Mathematical manual of the fitting engine

This manual documents, in a self-contained way, the complete mathematics of the fitting engine for ^{57}Fe Mössbauer spectra implemented in the application. It covers (i) the *forward model* —line profiles, positions and intensities of the six lines of the sextet with first-order and Kündig treatments of the magnetic–quadrupole coupling, doublets and singlets—; (ii) the *discrete fit* —data statistics in Gauss and Poisson modes, propagation of the calibration uncertainty, cost functional with robust losses, trust-region optimization with multistart and a global option, covariance, diagnostics and bootstrap—; and (iii) the *distribution fit* $P(B_{\mathrm{hf}})$ or $P(\Delta E_Q)$ —discretization, kernel, Tikhonov and total-variation regularization, conditioning, effective degrees of freedom, automatic selection of the regularization parameter, error bars and parametric distributions. References to the file and function of the code are included.

A.1 Forward model

A.1.1 Transmission spectrum

After folding the raw data around the fold point, the spectrum is defined over N channels with velocities $\{v_i\}$ and normalized transmission $\{y_i\}$, with $y_i \approx 1$ away from the absorption peaks. The forward model is a linear superposition of absorptions on top of a baseline:

$$\boxed{T(v;\boldsymbol{\theta}) \;=\; b + s\,v \;-\; \sum_{c=1}^{N_c} A_c(v;\boldsymbol{\theta}_c)} \tag{A.1}$$

where b is the baseline (multiplicative; the data are normalized by the factor `norm_factor`), s the slope, and each A_c is the absorption of a component. Besides sextet, doublet and singlet, the components can be **magnetic relaxation** models (phenomenological relaxation, Blume–Tjon, Néel; see the relaxation document). Implemented in `core/physics.py`, function `total_model`.

By default it is a *thin-absorber* model (linear in depth). Optionally a **thick-absorber** model with saturation is applied, $T = b+s\,v-C\big(1-e^{-A_{\mathrm{tot}}/C}\big)$ (parameter $C =$ `sat_scale`); the derivation is in the thickness-correction document.

Drive waveform and velocity axis

With a **constant-acceleration** drive (triangular wave) the velocity is linear in the channel: after folding, $\{v_i\}$ is **equispaced** on $[-v_{\max}, v_{\max}]$ (`velocity_axis` in `core/folding.py`). With a **sinusoidal** drive the velocity is not linear; following the NORMOS criterion (`TRIANG=.FALSE.`, `FOLD=.FALSE.`, `SIMULT=.TRUE.`) the data are *not folded* and each channel is assigned its actual velocity

$$v_i \;=\; v_{\max}\,\sin\!\left(\frac{2\pi\,(i-c_0)}{N}\right), \tag{A.2}$$

with c_0 the phase (channel of $v=0$), auto-detected by symmetry and refinable as a fit parameter. The resulting axis is not monotonic, which the model tolerates because it is evaluated point by point (`sine_velocity_axis` in `core/folding.py`).

A.1.2 Line profiles

Each elementary line uses a profile normalized to unit height at its center. The application offers two profiles, selectable globally.

Lorentzian

$$L(v;v_0,\gamma) \;=\; \frac{(\gamma/2)^2}{(v-v_0)^2+(\gamma/2)^2}, \tag{A.3}$$

where γ is the **width parameter that is fitted**: the *full width at half maximum* (FWHM). The half-width (HWHM) is therefore $\gamma/2$, which is what appears in the formula. It satisfies $L(v_0)=1$.

Width convention

In the code and in the interface, the width parameter (γ_1, etc.) is the **FWHM**. Internally the profile uses the half-width $\gamma/2$ (`lorentzian` in `core/physics.py` divides by two before evaluating). Reading γ as HWHM (as in old versions of this document) gave widths off by a factor of 2.

Pseudo-Voigt (Faddeeva)

The convolution of a Lorentzian (HWHM $\gamma/2$) with a Gaussian (width σ) is the Voigt profile, evaluated through the Faddeeva function $w(z)=e^{-z^2}\,\mathrm{erfc}(-iz)$:

$$\widetilde{V}(v;v_0,\gamma,\sigma) \;=\; \frac{1}{\sigma\sqrt{2\pi}}\,\Re\!\left[w\!\left(\frac{(v-v_0)+i\gamma/2}{\sigma\sqrt{2}}\right)\right]. \tag{A.4}$$

So that the profile has height exactly 1 at v_0 *independently of the sampling*, it is normalized by its analytical value at the peak:

$$V(v;v_0,\gamma,\sigma) \;=\; \frac{\widetilde{V}(v;v_0,\gamma,\sigma)}{\widetilde{V}(v_0;v_0,\gamma,\sigma)}, \qquad \widetilde{V}(v_0)=\frac{1}{\sigma\sqrt{2\pi}}\,\Re\!\left[w\!\left(\tfrac{i\gamma/2}{\sigma\sqrt{2}}\right)\right]. \tag{A.5}$$

Since $w(iy)=e^{y^2}\,\mathrm{erfc}(y)\in\mathbb{R}^+$ for $y\in\mathbb{R}$, the denominator is real and positive. In the limit $\sigma\to 0$, $V\to L$.

Normalization to the peak

Normalizing by the *discrete* maximum on the grid (as previous versions did) underestimates the true peak when no v_i falls at v_0, inflating the amplitude and making the depth depend on the sampling. Form (A.5) removes that bias.

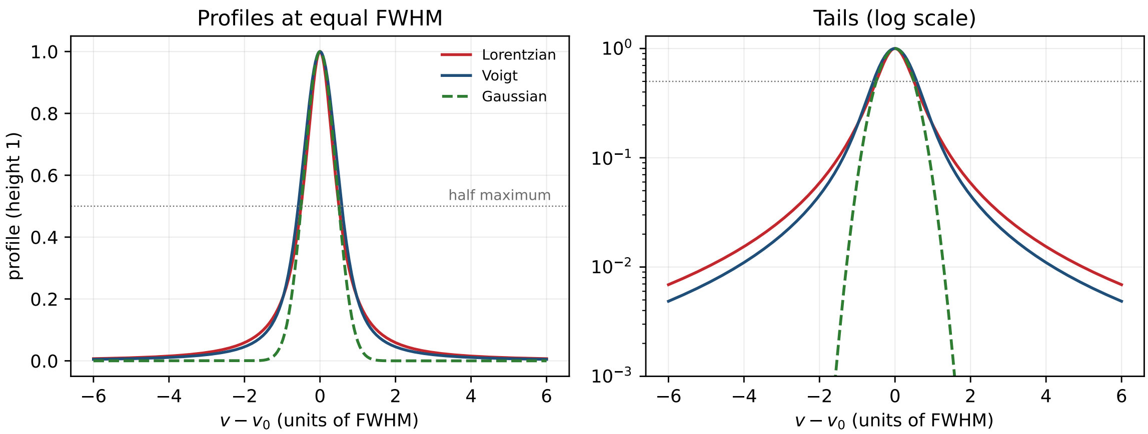

Figure A.1: Line profiles normalized to unit height and to equal FWHM. The Lorentzian (A.3) has *heavy tails*; the Gaussian decays much faster; the pseudo-Voigt (A.5) interpolates between the two. The logarithmic scale (right) makes the weight of the tails visible, which is what distinguishes one profile from another far from the center.

A.1.3 Magnetic sextet

A sextet arises from the nuclear Zeeman splitting combined, where applicable, with the quadrupole interaction. The six lines correspond to the allowed transitions ($\Delta m = 0, \pm 1$) between the ground state $I_g = \frac{1}{2}$ and the excited state $I_e = \frac{3}{2}$. The absorption is

$$A_{\text{sext}}(v) \;=\; d \sum_{j=1}^{6} W_j \, P(v; v_{0,j}, \gamma_j)\,, \tag{A.6}$$

with d the depth, P the profile (§A.1.2), W_j the intensities (§A.1.4) and γ_j the widths

$$\boldsymbol{\gamma} = \gamma_1\,[1,\ \gamma_2,\ \gamma_3,\ \gamma_3,\ \gamma_2,\ 1], \tag{A.7}$$

where γ_2, γ_3 are *relative* widths. The positions $v_{0,j}$ are obtained by one of two treatments. Figure A.2 summarizes the physical origin of the six lines.

First-order treatment

The quadrupole is treated as a rigid perturbation on the Zeeman pattern:

$$v_{0,j} \;=\; \frac{B_{\text{hf}}}{B_0}\, r_j^{(0)} \;+\; \delta \;+\; s_j\, \Delta E_Q, \qquad j = 1, \ldots, 6, \tag{A.8}$$

with $B_0 = 33\,\text{T}$ the reference field, δ the isomer shift, $r^{(0)}$ the positions calibrated at B_0 (Appendix A.4.1) and the quadrupole pattern

$$\boldsymbol{s} = \left[\,+\tfrac{1}{2},\ -\tfrac{1}{2},\ -\tfrac{1}{2},\ -\tfrac{1}{2},\ -\tfrac{1}{2},\ +\tfrac{1}{2}\,\right]. \tag{A.9}$$

Valid when $\Delta E_Q \ll \mu_N g_e B_{\text{hf}}$ and the axis of the electric field gradient (EFG) is parallel to $\vec{B}$.

Kündig treatment (full axial Hamiltonian)

For an axial EFG (asymmetry parameter $\eta = 0$) and arbitrary angle β between $\vec{B}$ and the principal axis V_{zz}, the positions are obtained by diagonalizing the excited-state Hamiltonian. In the basis $\{|\frac{3}{2}\rangle, |\frac{1}{2}\rangle, |-\frac{1}{2}\rangle, |-\frac{3}{2}\rangle\}$:

$$\mathcal{H}_e \;=\; \omega_e\, I_z \;+\; \frac{\Delta E_Q}{6}\big(3I_{z'}^2 - I(I+1)\big), \qquad I_{z'} = I_z \cos\beta + I_x \sin\beta, \tag{A.10}$$

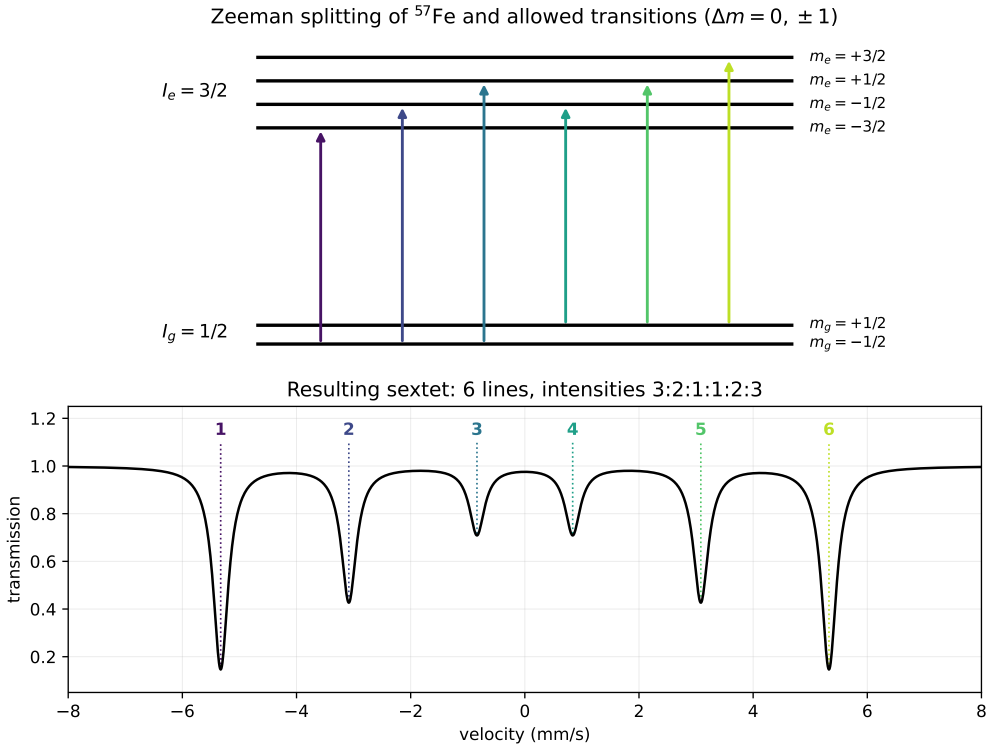

Figure A.2: Origin of the sextet. The hyperfine field Zeeman-splits the ground level ($I_g = \frac{1}{2}$, two sublevels) and the excited level ($I_e = \frac{3}{2}$, four sublevels). The six allowed transitions ($\Delta m = 0, \pm 1$) produce the six lines, with intensities $3:2:1:1:2:3$ in a powder and positions at 33 T of $\pm 0{,}839/\pm 3{,}084/\pm 5{,}329$ mm/s. The quadrupole (not shown) rigidly shifts this pattern according to (A.8).

Figure A.3: First-order effect of the quadrupole (A.8). Compared with the pure sextet ($\Delta E_Q = 0$, gray), a $\Delta E_Q \neq 0$ shifts the lines according to the pattern $\boldsymbol{s} = [+\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}]$: the two outer lines move one way and the four inner lines the other, breaking the symmetry of the sextet. The magnitude of the shift measures ΔE_Q.

with $I(I+1)=\frac{15}{4}$ and $\omega_e=(B_{\rm hf}/B_0)\,\omega_e^{(0)}$ the Zeeman splitting of the excited state. Numerical diagonalization of the 4×4 matrix gives the eigenvalues E^e_m and eigenvectors. The ground state ($I_g=\frac{1}{2}$) is pure Zeeman:

$$E^g_{\pm1/2}=\pm\tfrac{1}{2}\,\omega_g,\qquad \omega_g=(B_{\rm hf}/B_0)\,\omega_g^{(0)}. \tag{A.11}$$

The six positions are the energy differences of the allowed transitions, assigning to each excited-state eigenvalue its dominant m_e (by eigenvector overlap):

$$\begin{aligned} v_{0,1}&=E^e_{-3/2}-E^g_{-1/2}, & v_{0,2}&=E^e_{-1/2}-E^g_{-1/2}, & v_{0,3}&=E^e_{+1/2}-E^g_{-1/2},\\ v_{0,4}&=E^e_{-1/2}-E^g_{+1/2}, & v_{0,5}&=E^e_{+1/2}-E^g_{+1/2}, & v_{0,6}&=E^e_{+3/2}-E^g_{+1/2}, \end{aligned} \tag{A.12}$$

plus δ. In the limit $\beta=0$ it reproduces (A.8) to machine precision. Implemented in `core/hamiltonian.py`.

Polycrystalline average

In a powder sample all orientations β are equiprobable. The absorption is averaged over the sphere:

$$A^{\rm pol}_{\rm sext}(v)\;=\;\frac{1}{2}\int_0^{\pi}\sin\beta\;A_{\rm sext}(v;\beta)\,d\beta\;\approx\;\sum_{k=1}^{N_q}w_k\,A_{\rm sext}(v;\beta_k), \tag{A.13}$$

with Gauss–Legendre quadrature nodes and weights over $x=\cos\beta\in[-1,1]$, $\beta_k=\arccos x_k$, $w_k=\frac{1}{2}w_k^{\rm GL}$ (so that $\sum_k w_k=1$). The computation is vectorized: for N_q orientations a tensor $(N_q,4,4)$ is built and diagonalized at once. Default value $N_q=20$ (error $<0{,}01\%$).

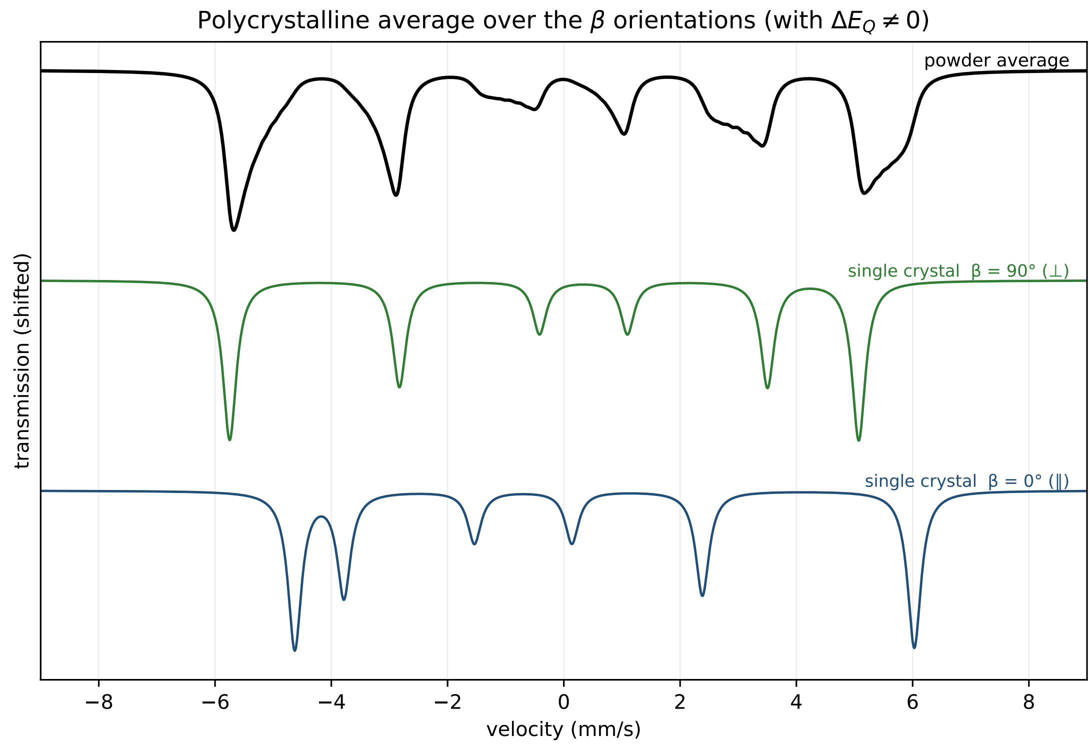

Figure A.4: Polycrystalline average when $\Delta E_Q\neq0$. In a single crystal the positions of the six lines depend on the angle β between $\vec{B}$ and the EFG axis (curves $\beta=0^\circ$ and $\beta=90^\circ$, computed with the Kündig Hamiltonian (A.10)). In a powder, all orientations coexist: the spectrum is the weighted sum (A.13) over β, which *broadens* and deforms the lines (upper curve). This is why the correct treatment of a quadrupolar sextet in a powder requires averaging, not a single orientation.

A.1.4 Intensities and texture

The intensities of the six lines respect the symmetry $1 \leftrightarrow 6$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$:

$$\boldsymbol{W} = [\, I_1,\, I_2,\, I_3,\, I_3,\, I_2,\, I_1\,], \qquad I_1 = i_3 i_1, \quad I_2 = i_3 i_2, \quad I_3 = i_3. \tag{A.14}$$

The application offers two parametrizations per sextet.

Free intensities

i_1, i_2, i_3 are independent fit parameters. It does not impose the 3:2:1 ratio; useful for numerical flexibility and empirical interpretation.

Texture parameter

For a sample with angle θ between the magnetic moment and the direction of the γ photon, the magnetic-dipole amplitudes are

$$W_{1,6} \propto 3(1 + \cos^2\theta), \quad W_{2,5} \propto 4\sin^2\theta, \quad W_{3,4} \propto (1 + \cos^2\theta). \tag{A.15}$$

Defining the texture parameter $t = \sin^2\theta \in [0, 1]$ and normalizing by $(1 + \cos^2\theta) = 2 - t$:

$$\boxed{i_1 = 3, \qquad i_2 = \frac{4t}{2-t}, \qquad i_3 = 1.} \tag{A.16}$$

Cases: $t = \frac{2}{3} \Rightarrow$ 3:2:1 (random powder); $t = 1 \Rightarrow$ 3:4:1 (flat texture, foil $\perp$ beam); $t = 0 \Rightarrow$ 3:0:1 (easy axis $\parallel$ beam). In this mode t is the only free intensity parameter; i_1, i_2, i_3 are derived from (A.16).

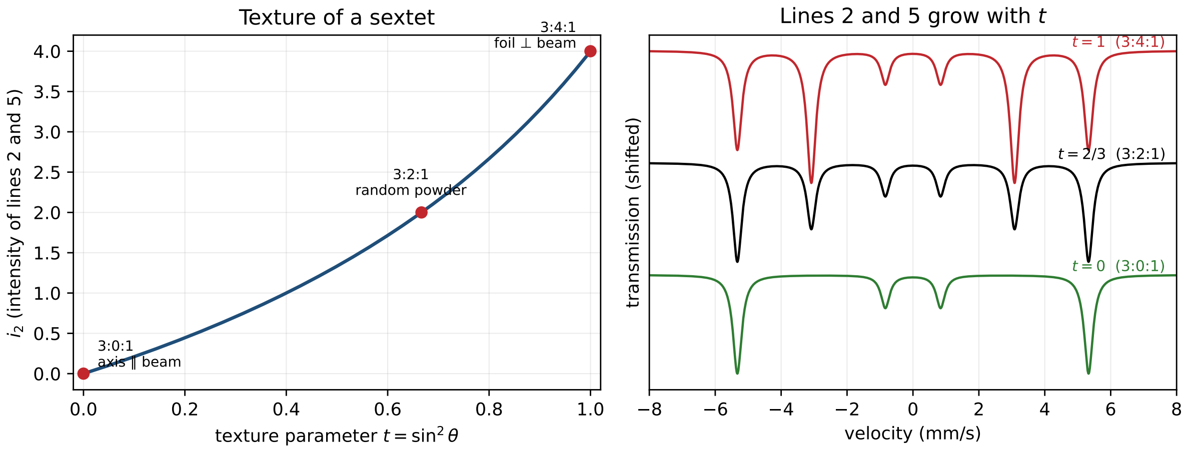

Figure A.5: Sextet texture. *Left*: the intensity of the central lines 2 and 5, $i_2 = 4t/(2-t)$ with $t = \sin^2\theta$ (A.16), varies between 0 (easy axis parallel to the beam, 3:0:1) and 4 (perpendicular foil, 3:4:1), passing through the random powder 3:2:1. *Right*: the three situations on the same sextet; lines 2 and 5 grow with t while the others do not change.

A.1.5 Doublet and singlet

$$A_{\text{dob}}(v) = d\big[\, i_1\, P(v; \delta - \tfrac{\Delta E_Q}{2}, \gamma_1) + i_1 i_2\, P(v; \delta + \tfrac{\Delta E_Q}{2}, \gamma_1\gamma_2)\,\big], \tag{A.17}$$
$$A_{\text{sing}}(v) = d\, i_1\, P(v; \delta, \gamma_1). \tag{A.18}$$

A.2 Discrete fit

A.2.1 Parametrization and constraints

The vector of free parameters $\boldsymbol{x} \in \mathbb{R}^p$ is composed of the globals $\{b, s\}$ (plus optionally $v_{\max}$ and the fold center c_{fold}) and, for each active component, a subset of $\boldsymbol{\theta}_c = (\delta, \Delta E_Q, B_{\text{hf}}, \gamma_1, \gamma_2, \gamma_3, d, i_1, i_2, i_3, t, \beta)$ according to its type and mode:

- *Type*: a singlet uses $\{\delta, \gamma_1, d, i_1\}$; a doublet adds $\{\Delta E_Q, \gamma_2, i_2\}$; a sextet uses all of them.
- *Intensities*: in texture mode, $\{i_1, i_2, i_3\}$ are replaced by the single parameter t.
- *Quadrupole*: in Kündig mode with fixed β, β enters as a free parameter; in polycrystal or first-order mode, it does not.
- *Fixed*: flagged parameters are removed from the active vector.
- *Linearly constrained* $k_t = \alpha\, k_s + \beta_0$: applied at each evaluation (up to six iterations for chains of dependencies).

Each parameter has hard bounds $[\ell_k, u_k]$ (Appendix A.4.2).

A.2.2 Data statistics

Gauss mode

Folding averages two independent Poisson channels; for counts c_i per folded channel, $\text{Var} \approx c_i/2$, from which the 1σ uncertainty of the normalized datum is:

$$\sigma_i \;=\; \frac{\max\!\big(\sqrt{c_i/2},\, 1\big)}{f_{\text{norm}}}, \qquad \sigma_i \leftarrow \max(\sigma_i, 10^{-9}). \tag{A.19}$$

Poisson mode (IRLS)

The Poisson likelihood is approximated by iteratively reweighted least squares: the uncertainty is evaluated on the *model* (predicted counts $\hat{c}_i = T_i f_{\text{norm}}$) and recomputed at each evaluation of the optimizer, which is equivalent to a Newton step on the Poisson $-2\log\mathcal{L}$:

$$\sigma_i^{\text{Poiss}} \;=\; \frac{\sqrt{\max(\hat{c}_i/2,\, 1)}}{f_{\text{norm}}} \;=\; \frac{\sqrt{\max(T_i f_{\text{norm}}/2,\, 1)}}{f_{\text{norm}}}. \tag{A.20}$$

Propagation of the calibration uncertainty

If the velocity calibration contributes an uncertainty $\sigma_{v_{\max}}$, its effect on the model is added in quadrature. Since $v_i = v_{\max}(2i/(N-1)-1)$, we have $\partial v_i/\partial v_{\max} = v_i/v_{\max}$, and by the chain rule

$$\sigma_i^{\text{eff}\,2} \;=\; \sigma_i^2 \;+\; \left(\frac{\partial T}{\partial v}\bigg|_i \frac{v_i}{v_{\max}}\,\sigma_{v_{\max}}\right)^2. \tag{A.21}$$

The term weighs more on the flanks of the peaks (where $\partial T/\partial v$ is large), which is where a scale error biases δ and B_{hf}.

A.2.3 Cost functional and robust losses

The weighted residual is

$$r_i(\boldsymbol{x}) \;=\; \frac{T(v_i;\boldsymbol{x}) - y_i}{\sigma_i^{\mathrm{eff}}}, \tag{A.22}$$

and the minimized functional

$$\mathcal{C}(\boldsymbol{x}) \;=\; \tfrac{1}{2}\sum_{i=1}^{N} \rho\big(r_i(\boldsymbol{x})^2\big)\,, \tag{A.23}$$

with ρ the loss function: *linear* $\rho(z) = z$ (least squares), *soft-*ℓ_1 $\rho(z) = 2(\sqrt{1+z} - 1)$, or *Huber*. The robust ones use scale $f_{\mathrm{scale}} = 3$ (threshold $\sim 3\sigma$), reducing the weight of channels with spurious peaks. For linear ρ, $\mathcal{C} = \frac{1}{2}\chi^2$. Figure A.6 compares the losses and the weight each one assigns.

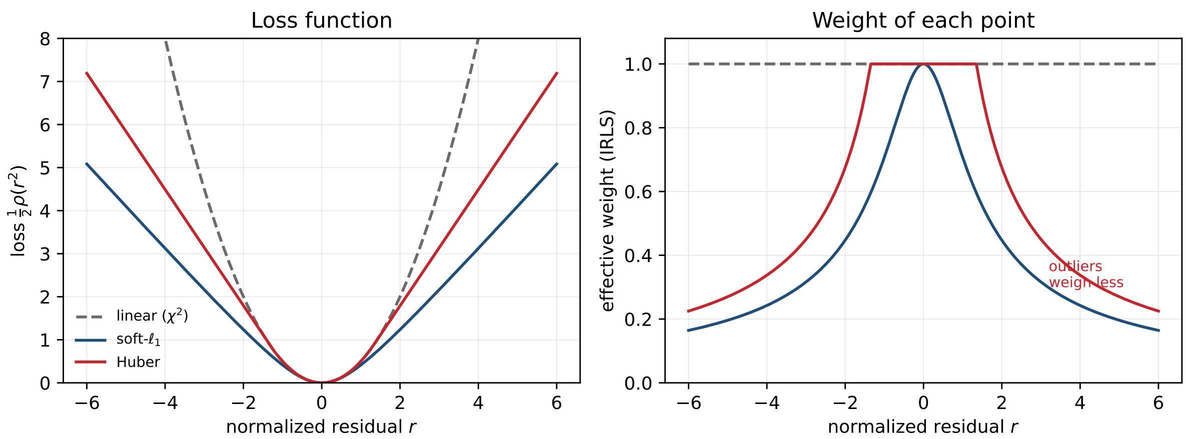

Figure A.6: Robust losses versus least squares. *Left*: the linear loss (χ^2) grows as r^2, while soft-ℓ_1 and Huber grow *linearly* for large residuals. *Right*: the effective weight (IRLS reweighting) is constant for the linear loss, but decays with $|r|$ in the robust ones, so that an outlier channel (spurious peak, cosmic ray) influences the fit much less.

A.2.4 Optimization

Trust region (TRF)

The minimization uses *Trust Region Reflective* over the bounds:

$$\hat{\boldsymbol{x}} \;=\; \operatorname*{arg\,min}_{\boldsymbol{\ell}\le\boldsymbol{x}\le\boldsymbol{u}} \;\mathcal{C}(\boldsymbol{x}), \tag{A.24}$$

solving at each iteration the quadratic subproblem with a finite-difference Jacobian:

$$\min_{\boldsymbol{d}} \;\tfrac{1}{2}\|J_k\boldsymbol{d} + \boldsymbol{r}_k\|^2 \;\text{ s.t. }\; \|\boldsymbol{D}_k\boldsymbol{d}\| \le \Delta_k, \;\; \boldsymbol{\ell} - \boldsymbol{x}_k \le \boldsymbol{d} \le \boldsymbol{u} - \boldsymbol{x}_k. \tag{A.25}$$

Multistart

The user's point plus 8 Gaussian restarts are tried:

$$\boldsymbol{x}^{(s)} = \boldsymbol{x}^{(0)} + \boldsymbol{\xi}^{(s)}, \quad \xi_k^{(s)} \sim \mathcal{N}(0, \rho_k^2(u_k - \ell_k)^2), \tag{A.26}$$

with $\rho_k = 0{,}12$ for $\{\delta, \Delta E_Q, B_{\mathrm{hf}}, \gamma_1, d, v_{\max}\}$ and 0,08 for the rest; fixed seed. The one with the lowest cost is retained.

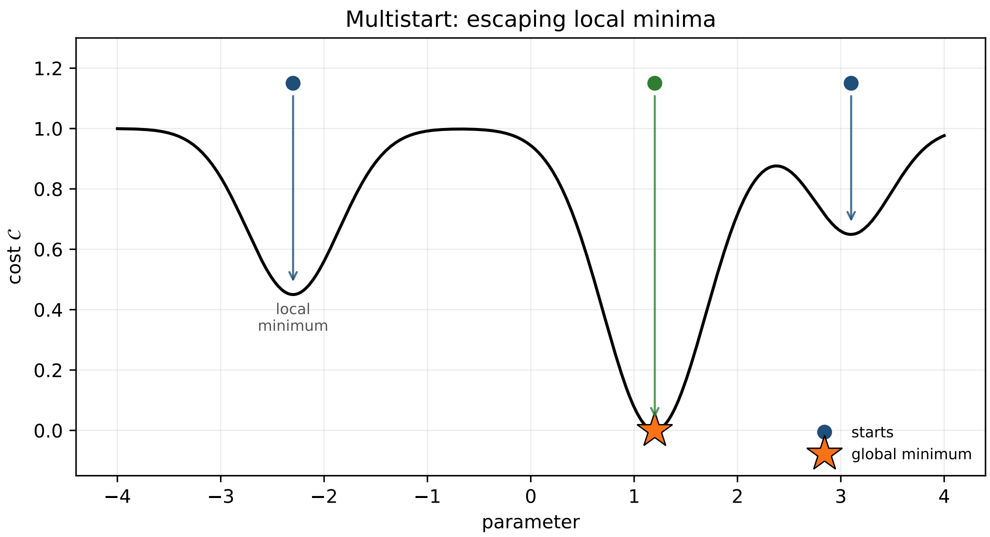

Figure A.7: Multistart. The cost may have local minima in addition to the global one. Starting from several initial points and keeping the one with the lowest cost reduces the risk of converging to a local minimum: in the example, only the central start reaches the global minimum.

Global optimization (optional)

As an optional pre-pass, *differential evolution* is run on the scalar cost $\frac{1}{2}\|\boldsymbol{r}\|^2$ (Sobol initialization, no polishing), and its best point is added as a seed for the TRF polishing. It covers local minima due to label permutation and $B_{\text{hf}}^{(1)} \leftrightarrow B_{\text{hf}}^{(2)}$ swaps in multi-sextet spectra that the Gaussian start does not reach.

A.2.5 Uncertainties and diagnostics

Covariance

From the final Jacobian $J = U\Sigma V^\top$ (SVD, truncated singular values):

$$\boxed{\widehat{\mathrm{Cov}}(\hat{\boldsymbol{x}}) = V\,\Sigma^{-2}V^\top \cdot \frac{2\,\mathcal{C}(\hat{\boldsymbol{x}})}{N-p}, \qquad \hat{\sigma}_{x_k} = \sqrt{\widehat{\mathrm{Cov}}_{kk}}.} \tag{A.27}$$

Correlations $|\rho_{kl}| \geq 0{,}95$ are reported as a warning of degeneracy.

Goodness-of-fit statistics

$$\chi^2 = \sum_i r_i^2, \qquad \nu = N - p, \qquad \chi^2_\nu = \chi^2/\nu, \tag{A.28}$$

$$\mathrm{AIC} = N\ln(\mathrm{RSS}/N) + 2p, \qquad \mathrm{BIC} = N\ln(\mathrm{RSS}/N) + p\ln N. \tag{A.29}$$

Residual diagnostics

To detect non-random structure (with $\tilde{r} = r - \bar{r}$):

$$\text{lag-1 autocorrelation:} \quad \rho_1 = \frac{\sum_i \tilde{r}_i\tilde{r}_{i+1}}{\sum_i \tilde{r}_i^2}, \tag{A.30}$$

$$\text{runs (Wald–Wolfowitz):} \quad Z = \frac{R - \mu_R}{\sigma_R}, \tag{A.31}$$

$$\text{antisymmetric correlation:} \quad c_{\text{as}} = \frac{\tilde{\boldsymbol{r}} \cdot (-\tilde{\boldsymbol{r}}^{\text{rev}})}{\tilde{\boldsymbol{r}} \cdot \tilde{\boldsymbol{r}}}, \tag{A.32}$$

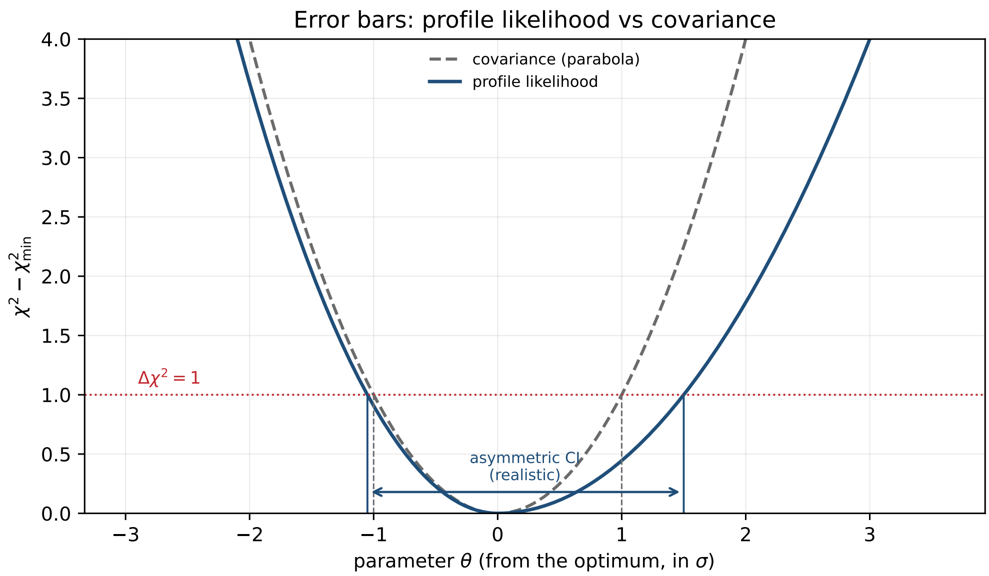

Figure A.8: Covariance versus profiled likelihood. The covariance (A.27) approximates the χ^2 well by a parabola and gives a *symmetric* error. The profiled likelihood scans the parameter while refitting the others and measures the real χ^2; the cut $\Delta\chi^2 = 1$ gives an *asymmetric* confidence interval, more realistic when the well is not parabolic.

with warnings if $|\rho_1| > 0{,}35$, $|Z| > 2$ or $c_{\text{as}} > 0{,}45$ (this last one detects centering/calibration errors).

Bootstrap

Monte Carlo estimation of errors: M replicas of synthetic data $\tilde{y}^{(m)} = T(\cdot;\hat{\boldsymbol{x}}) + \eta^{(m)}$, with $\eta^{(m)}$ Gaussian of standard deviation σ_i (or actual Poisson resampling $\tilde{c} \sim \text{Poisson}(\hat{c})$ in Poisson mode); each replica is refitted and $\hat{\sigma}^{\text{boot}}_{x_k} = \text{std}_m(\hat{x}_k^{(m)})$.

A.3 Distribution fit

A.3.1 Discretization and kernel

The spectrum is modeled as a continuous density $P(B_{\text{hf}})$ (or $P(\Delta E_Q)$) discretized into n *bins* with equispaced centers $c_k \in [p_{\min}, p_{\max}]$. The *kernel*

$$K_{ik} \;=\; A^{(d=1)}_{\text{sext}}\big(v_i;\; B_{\text{hf}} = c_k\big) \tag{A.33}$$

is the absorption at v_i of a sextet of unit depth with field c_k (variable B_{hf}) or with $\Delta E_Q = c_k$ and fixed B_{hf} (variable ΔE_Q). The kernel intensities follow 3:2:1 by construction.

A.3.2 Augmented linear system

The forward model is

$$\hat{y}_i \;=\; b + s\,v_i \;-\; \sum_k K_{ik}\,P_k \;-\; \sum_m K^{\text{sharp}}_{im} q_m, \tag{A.34}$$

with $P_k \geq 0$ the distribution weights and $q_m \geq 0$ amplitudes of optional "sharp" sextets. Defining the design matrix

$$X = \big[\,\mathbf{1}\;\big|\;\boldsymbol{v}\;\big|\;-K\;\big|\;-K^{\text{sharp}}\,\big], \qquad \boldsymbol{z} = (b, s, \boldsymbol{P}^\top, \boldsymbol{q}^\top)^\top, \tag{A.35}$$

and the weights $W = \mathrm{diag}(1/\sigma_i^2)$, the problem is

$$\boxed{\min_{\boldsymbol{z}} \; \left\|W^{1/2}(X\boldsymbol{z} - \boldsymbol{y})\right\|_2^2 \; + \; \alpha\,\mathcal{R}(\boldsymbol{P}) \quad \text{s.t.} \quad b \geq 0, \; \boldsymbol{P} \geq 0, \; \boldsymbol{q} \geq 0.} \tag{A.36}$$

The slope s has no bound. It is solved with bounded least squares (`lsq_linear`, reflective Newton with LSMR).

Column-scale preconditioning

For small B_{hf} the sextet collapses and the columns of K become almost collinear with widely disparate norms, which ill-conditions (A.36). Each distribution column is rescaled $K_k \to K_k/s_k$ with $s_k = \|W^{1/2}K_k\|$ and $\tilde{P}_k = s_k P_k$; the penalizer and the weights are adjusted accordingly and the scale is undone at the end. It is an *exact* reparametrization (solution and degrees of freedom invariant) that only improves numerical stability.

A.3.3 Regularization

The inverse problem is *ill-conditioned*: many different $\boldsymbol{P}$ reproduce the spectrum almost equally well. Adding a term $\alpha\,\mathcal{R}(\boldsymbol{P})$ selects a stable solution. The value of α is the trade-off between fidelity and smoothness; figure A.9 shows the three regimes.

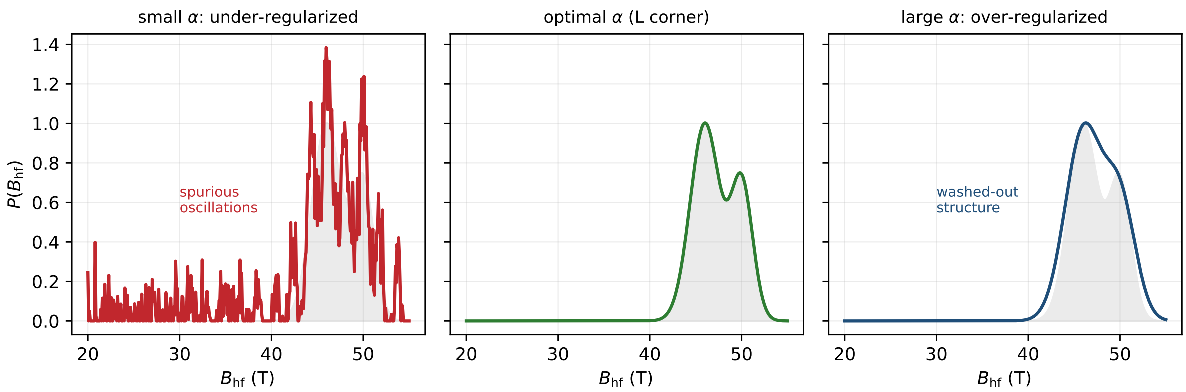

Figure A.9: Effect of the regularization parameter α on the distribution recovered from the same spectrum. With α too small (under-regularized), *spurious oscillations* appear that fit the noise; with α too large (over-regularized) the real structure is *washed out*; the value at the corner of the L-curve (see the distributions chapter of the user manual) maintains the balance. In gray, the true distribution.

Tikhonov (smoothness, L2)

$\mathcal{R}(\boldsymbol{P}) = \|L\,\boldsymbol{P}\|_2^2$ with L the *second-difference* operator $(n-2)\times n$:

$$(L\boldsymbol{P})_k = P_{k-1} - 2P_k + P_{k+1}, \tag{A.37}$$

which penalizes curvature and produces smooth distributions. The augmented system stacks $\left[\begin{smallmatrix} W^{1/2}X \\ \sqrt{\alpha}\,L_{\mathrm{ext}} \end{smallmatrix}\right]\boldsymbol{z} = \left[\begin{smallmatrix} W^{1/2}\boldsymbol{y} \\ \boldsymbol{0} \end{smallmatrix}\right]$.

Total variation (edges, L1)

$\mathcal{R}(\boldsymbol{P}) = \|D\,\boldsymbol{P}\|_1$ with D the *first difference* $(D\boldsymbol{P})_k = P_{k+1} - P_k$. It favors piecewise-constant solutions that preserve edges (less blurring of peaks). It is solved by iteratively reweighted least squares (IRLS): starting from the majorization $|t| \le t^2/(2|t^{\text{prev}}|) + \text{const}$, each iteration solves a weighted Tikhonov with

$$w_k = \frac{1}{\sqrt{(D\boldsymbol{P}^{\text{prev}})_k^2 + \varepsilon^2}}, \qquad \mathcal{R} \approx \sum_k w_k \, (D\boldsymbol{P})_k^2, \tag{A.38}$$

reusing `lsq_linear`. The effective operator $\sqrt{w}\,D$ is also used for the degrees of freedom and the error bars.

Maximum entropy (MaxEnt)

Third option of `reg_mode`. Instead of penalizing curvature or edges, it maximizes the *entropy* of the distribution (the most "uniform" solution compatible with the data, without introducing unjustified structure):

$$\min_{\boldsymbol{z}} \;\; \tfrac{1}{2}\big\|(\boldsymbol{y} - X\boldsymbol{z})/\boldsymbol{\sigma}\big\|_2^2 \;-\; \alpha\, S(\boldsymbol{P}), \qquad S(\boldsymbol{P}) = -\sum_k P_k \,\log\frac{P_k}{m_k}, \tag{A.39}$$

with m_k a default model (uniform). Since the entropy is **not quadratic**, `lsq_linear` cannot be used: it is optimized with **L-BFGS-B** and an analytical gradient over the physical variables $\boldsymbol{z} = [b, s, \boldsymbol{P}, \boldsymbol{q}]$ (`_solve_maxent` in `mossbauer_distribution.py`). The effective degrees of freedom use the Hessian corresponding to this functional.

A.3.4 Poisson likelihood in the distribution

In Poisson mode the histogram fit is repeated with IRLS reweighting of σ from the model (§A.2.2), closing the loop through `lsq_linear` until convergence.

A.3.5 Effective degrees of freedom

Regularization reduces the degrees of freedom below n. The effective number is the trace of the hat matrix:

$$p_{\text{eff}} \;=\; \operatorname{tr} A(\alpha), \qquad A(\alpha) = X\big(X^\top W X + \alpha L^\top L\big)^{-1} X^\top W, \tag{A.40}$$

computable as $\operatorname{tr}\!\big[(X^\top W X + \alpha L^\top L)^{-1} X^\top W X\big]$. It is used in $\chi^2_\nu = \chi^2/(N - p_{\text{eff}})$, avoiding the bias of counting the n bins as independent parameters. It is invariant under the column rescaling of §A.3.2.

A.3.6 Selection of the regularization parameter

Dimensionless α

The α set by the user is **not** the raw weight of (A.36). The data term is a $\chi^2 \sim N$, but the penalizer is in units of absorption; without normalization, the "useful" α would depend on N, on the noise σ and on the absorption depth, and the same number would mean different things in each spectrum. That is why the interface's α is **made dimensionless** before applying it:

$$\alpha_{\text{eff}} \;=\; \alpha\, \lambda_{\text{ref}}\, c_0, \qquad \lambda_{\text{ref}} = \frac{\|A_{\text{dist}}\|_F^2}{\|L_{\text{scaled}}\|_F^2}, \qquad c_0 = 10^{-4}, \tag{A.41}$$

where A_{dist} and L_{scaled} are the distribution part of $W^{1/2}X$ and the *whitened* penalizer (after the column rescaling of §A.3.2), $\|\cdot\|_F$ the Frobenius norm and $c_0 =$ `ALPHA_REF_SCALE`. The augmented system uses $\sqrt{\alpha_{\text{eff}}}\,L$ instead of $\sqrt{\alpha}\,L$; the same for p_{eff} (A.40) and the error bars. The effect: $\log_{10}\alpha \approx 0$ is the **natural balance** data/smoothness for *any* spectrum, and the elbow of the L-curve stays stable against N, noise and depth (typical useful values $-1 \ldots +2$). In 2D the same is done for each axis (α_B, α_Q) with its own λ_{ref}.

Three criteria are computed over a logarithmic grid of α.

L-curve (maximum curvature). With $\rho(\alpha) = \log\|W^{1/2}(X\hat{z} - y)\|$ and $\eta(\alpha) = \log\|L\hat{\boldsymbol{P}}\|$, the corner maximizes

$$\kappa(\alpha) = \frac{\rho'\eta'' - \eta'\rho''}{\big((\rho')^2 + (\eta')^2\big)^{3/2}}. \tag{A.42}$$

Trade-off. Minimum normalized distance to the origin in the (η, ρ) plane.

Generalized cross-validation (GCV).

$$\mathrm{GCV}(\alpha) = \frac{\|W^{1/2}(\boldsymbol{y} - X\hat{z})\|^2}{\big(N - p_{\mathrm{eff}}(\alpha)\big)^2}, \qquad \hat{\alpha} = \arg\min_{\alpha} \mathrm{GCV}(\alpha), \tag{A.43}$$

which does not require knowing the noise scale and is optimal in prediction loss.

A.3.7 Error bars of $P(B_{\mathrm{hf}})$

For the regularized estimator $\hat{\boldsymbol{z}} = H^{-1}X^\top W\boldsymbol{y}$ with $H = X^\top WX + \alpha L^\top L$, the linearized covariance is

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{z}}) \;=\; H^{-1}\,(X^\top WX)\,H^{-1} \cdot \frac{\chi^2}{N - p_{\mathrm{eff}}}, \tag{A.44}$$

scaled by the effective χ^2_ν (same convention as the discrete fit). The square root of the diagonal corresponding to the bins gives σ_{P_k}. The band $P_k \pm \sigma_{P_k}$ is clipped to $P \geq 0$ when plotted (non-negativity makes it asymmetric in bins pinned to zero). In TV mode the effective operator $\sqrt{w}\,D$ is used instead of L.

A.3.8 Parametric distributions

As an alternative to the regularized histogram, closed forms with few parameters are fitted (non-linear least squares, no α):

$$\begin{aligned}
\text{Gaussian:}\quad & P_k \propto \exp\!\big(-\tfrac{1}{2}((c_k - \mu)/\varsigma)^2\big), && (b, s, A, \mu, \log\varsigma), && \text{(A.45)}\\
\text{VBF:}\quad & P_k = \sum_{j=1}^{N} A_j\,\mathcal{N}(c_k; \mu_j, \sigma_j), && (b, s, \{A_j, \mu_j, \log\sigma_j\}_{j=1}^{N}), && \text{(A.46)}\\
\text{Binomial:}\quad & P_k \propto \binom{n-1}{k-1} p^{k-1}(1-p)^{n-k}, && (b, s, A, \operatorname{logit} p), && \text{(A.47)}\\
\text{Fixed:}\quad & P_k = \phi_k \text{ (preloaded)}, && (b, s, A). && \text{(A.48)}
\end{aligned}$$

VBF (Voigt-Based Fitting, Rancourt–Ping 1991). It generalizes the Gaussian to a sum of N Gaussians ($N = 1\ldots6$) over a Voigt kernel, each one with amplitude A_j, mean μ_j and width σ_j (stored ordered by μ). Since the Gaussian kernel $\mathcal{N}$ is **normalized to unit area**, the coefficient A_j *is* the **area** of that component over the grid; its fraction $A_j/\sum_i A_i$ gives the **population** of each environment, which Fitbauer reports as "area %". The *Gaussian* form is simply the case $N = 1$. Appendix B develops the complete theory of the method (forward model, Voigt profile, parametrization and best practices).

A.3.9 Correlation $\delta(H)$ and $\Delta E_Q(H)$ in the kernel

By default all the subspectra of the kernel share the same δ and ΔE_Q, and only the field varies. Optionally δ and ΔE_Q are coupled to the bin field through two slopes (Le Caër–Dubois / Wivel–Mørup model):

$$\delta_k = \delta_0 + \kappa_\delta\,(c_k - H_{\mathrm{ref}}), \qquad \Delta E_{Q,k} = \Delta E_{Q,0} + \kappa_q\,(c_k - H_{\mathrm{ref}}), \tag{A.49}$$

with H_{ref} a reference field. Thus the kernel K_{ik} is built using $(\delta_k, \Delta E_{Q,k})$ in each column instead of global values. With $\kappa_\delta = \kappa_q = 0$ the classic case is recovered. The slopes are refined in the outer loop (§A.3.10) and apply to the Histogram and to the VBF.

A.3.10 Global refinement (VARPRO)

When enabled, a nonlinear outer loop optimizes the shape parameters $\boldsymbol{\eta} = (\delta_g, \log \gamma_g, \dots)$; at each evaluation the inner linear problem (A.36) is solved *in full*, returning the residual to the optimizer (separable VARPRO-type scheme):

$$\min_{\boldsymbol{\eta}} \left\| W^{1/2} \big[X(\boldsymbol{\eta})\, \hat{\boldsymbol{z}}(\boldsymbol{\eta}) - \boldsymbol{y} \big] \right\|^2, \qquad \hat{\boldsymbol{z}}(\boldsymbol{\eta}) = \arg\min_{\boldsymbol{z}} (\text{A.36}). \tag{A.50}$$

A.4 Appendices

A.4.1 Sextet calibration

The reference positions at $B_0 = 33{,}0\,\text{T}$ are the **published** ones of α-Fe (standard velocity reference), with no additional rescaling factor:

$$r^{(0)} = \tfrac{1}{2}[-10.657, -6.167, -1.677, 1.677, 6.167, 10.657] = [\mp 5{,}3285,\ \mp 3{,}0835,\ \mp 0{,}8385]\ \text{mm/s}. \tag{A.51}$$

The positions for a field B_{hf} scale linearly, $r^{(0)} \cdot (B_{\text{hf}}/33{,}0)$. From them the Zeeman splittings used in (A.10) are derived:

$$\omega_e^{(0)} = r_2^{(0)} - r_1^{(0)} \approx 2{,}245\ \text{mm/s}, \qquad \omega_g^{(0)} = -\big[(r_4^{(0)} - r_3^{(0)}) + \omega_e^{(0)}\big] \approx -3{,}922\ \text{mm/s}. \tag{A.52}$$

No factor 32.95

Previous versions of this document included a factor $(B_0/32{,}95)$ that over-scaled the positions by $\sim 0{,}15\%$. The code uses the published positions *as is* (`LINE_POS_33T` in `core/constants.py`), so that factor has been removed (and with it $\omega_e^{(0)}$, $\omega_g^{(0)}$ go from $2{,}248/-3{,}928$ to $2{,}245/-3{,}922$).

A.4.2 Parameter bounds

Parameter	Bound	Unit
b (baseline)	$[0{,}7,\ 1{,}3]$	–
s (slope)	$[-5\cdot 10^{-3},\ 5\cdot 10^{-3}]$	mm^{-1}s
δ	$[-2,\ 3]$	mm/s
ΔE_Q	$[-4,\ 4]$	mm/s
B_{hf}	$[0,\ 60]$	T
γ_1 (FWHM)	$[0{,}06,\ 4]$	mm/s
γ_2, γ_3	$[0{,}2,\ 6]$	(relative)
d (depth)	$[0,\ 0{,}30]$	–
i_1, i_2, i_3	$[0,9]/[0,6]/[0,3]$	(relative)
t (texture)	$[0,\ 1]$	–
β	$[0,\ 90]$	degrees

A.4.3 Faddeeva function

$w(z) = e^{-z^2}\,\text{erfc}(-iz)$, evaluated by `scipy.special.wofz`. The real part gives the Voigt profile (A.5); for $z = iy$ with $y \in \mathbb{R}$, $w(iy) = e^{y^2}\,\text{erfc}(y)$ is real, which allows the analytical normalization to the peak.

A.4.4 Summary of the optimization stack

Mode	**Functional**	**Solver**
Discrete	$\frac{1}{2}\sum_i \rho(r_i^2)$	`least_squares` (TRF) [+ optional DE]
Tikhonov distribution	$\lVert W^{1/2}(X\boldsymbol{z} - \boldsymbol{y})\rVert^2 + \alpha\lVert L\boldsymbol{P}\rVert^2$	`lsq_linear` (bounded)
TV distribution	$\lVert W^{1/2}(X\boldsymbol{z} - \boldsymbol{y})\rVert^2 + \alpha\lVert D\boldsymbol{P}\rVert_1$	IRLS over `lsq_linear`
Parametric distribution	$\lVert W^{1/2}(X(\boldsymbol{\theta})\boldsymbol{z} - \boldsymbol{y})\rVert^2$	`least_squares` (NL)
Global refinement	$\lVert W^{1/2}(X(\boldsymbol{\eta})\hat{\boldsymbol{z}}(\boldsymbol{\eta}) - \boldsymbol{y})\rVert^2$	outer NL + inner bounded

Appendix B

Voigt-based fitting (VBF): $P(B_{\rm hf})$ as a sum of Gaussians

B.1 Motivation: a basis of Gaussians for $P(B_{\rm hf})$

In a magnetically disordered solid —a metallic glass, a substituted ferrite, a nanoparticle, an amorphous material— the hyperfine field $B_{\rm hf}$ seen by each ^{57}Fe nucleus is not unique, but is distributed according to a probability density $P(B_{\rm hf})$. The measured spectrum is the *superposition* of all the elemental sextets weighted by that distribution. Recovering $P(B_{\rm hf})$ from the spectrum is an ill-conditioned inverse problem (see the appendix of the mathematical manual, §A.3.9 and nearby sections).

The Hesse–Rübartsch **histogram** solves it without assuming a shape: it assigns a free weight to each bin and controls the oscillations with a regularizer (Tikhonov, TV or MaxEnt) governed by α. It is flexible, but it delivers a profile without structure and is sensitive to the choice of α.

The **VBF** (*Voigt-Based Fitting*, Rancourt and Ping, 1991) adopts the opposite hypothesis, which is very reasonable in practice: each magnetic **environment** or **subpopulation** contributes a *Gaussian* of fields, and the total distribution is the sum of a few of them,

$$P(B_{\rm hf}) \;=\; \sum_{j=1}^{N} A_j\,\mathcal{N}\big(B_{\rm hf};\,\mu_j,\sigma_j\big), \qquad N = 1\ldots 6. \tag{B.1}$$

Instead of dozens of bin weights, the model has $3N$ shape parameters with **direct physical meaning**: the position μ_j (mean field of the environment), the width σ_j (its dispersion) and the area A_j (its population). By reducing the degrees of freedom so much, the fit is stable and **does not require** regularization or α.

B.2 From field to spectrum: the forward model

Fitbauer discretizes the variable over a grid of M nodes $c_1 < \cdots < c_M$ (the bin centers, `parameter_grid`). To each node c_k corresponds an **elemental subspectrum** —an Fe-57 sextet with field c_k, shift δ, quadrupole splitting ΔE_Q and width γ— whose absorption in the velocity channel v_i is the element of the **kernel**

$$K_{ik} \;=\; \sum_{\ell=1}^{6} I_\ell(c_k)\;\Phi\big(v_i - p_\ell(c_k);\,\Gamma_\ell,\varsigma\big), \tag{B.2}$$

where $p_\ell(c_k)$ and $I_\ell(c_k)$ are the positions and intensities of the six lines (appendix A) and Φ is the line profile (§B.3). Over this grid, equation (B.1) becomes the weight vector

$$w_k \;=\; \sum_{j=1}^{N} A_j\,\mathcal{N}(c_k;\mu_j,\sigma_j), \qquad k = 1,\ldots,M, \tag{B.3}$$

and the modeled spectrum is linear in those weights:

$$\hat{y}_i \;=\; b + s\,v_i \;-\; \sum_{k=1}^{M} K_{ik}\,w_k \;-\; \underbrace{\sum_m a_m\,S_{im}}_{\text{sharp components (opt.)}}, \tag{B.4}$$

with baseline b, slope s and, optionally, one or several *sharp* subspectra S_{im} (discrete crystalline phases) of amplitude $a_m \geq 0$.

Gaussian kernel normalized to unit area. The elemental Gaussian of (B.3) is evaluated and **normalized to unit area** over the grid itself,

$$\mathcal{N}(c_k;\mu,\sigma) \;=\; \frac{\exp\!\big(-\tfrac{1}{2}((c_k-\mu)/\sigma)^2\big)}{\displaystyle\int \exp\!\big(-\tfrac{1}{2}((c-\mu)/\sigma)^2\big)\,\mathrm{d}c}, \tag{B.5}$$

where the denominator is computed by the trapezoidal rule (`gauss_weights`). This normalization is what gives the coefficient A_j the meaning of *area* (§B.5).

B.3 Why "Voigt-based"

The name of the method comes from the line profile Φ. The intrinsic Mössbauer line is **Lorentzian** of natural width, but the instrumental broadening and the finite thickness of the absorber convolve it with a **Gaussian**. The convolution of the two is the **Voigt** profile

$$V(x;\gamma,\varsigma) \;=\; (L_\gamma * G_\varsigma)(x) \;=\; \frac{\mathrm{Re}\;\mathrm{w}(z)}{\varsigma\sqrt{2\pi}}, \qquad z = \frac{x + i\,\gamma/2}{\varsigma\sqrt{2}}, \tag{B.6}$$

with L_γ a Lorentzian of full width γ, G_ς a Gaussian of deviation ς (the *Voigt* σ parameter of the GUI) and $\mathrm{w}(z)$ the Faddeeva function. As $\varsigma \to 0$ the pure Lorentzian is recovered. In (B.2) one takes $\Phi = V$ ("Voigt" profile) or $\Phi = L$ ("Lorentz" profile); Fitbauer sets the profile explicitly when building the VBF kernel.

The dual reading of the Gaussian broadening. There is an important equivalence that justifies the approach. Distributing a *Lorentzian* sextet over a *Gaussian of fields* of width σ_j is equivalent, on the velocity axis, to convolving that sextet with a Gaussian whose width is σ_j scaled by the sensitivity $\partial p_\ell/\partial B_{\mathrm{hf}}$ of each line to the field. That is, **each component of $P(B_{\mathrm{hf}})$ produces a sextet of effectively *Voigtian* lines**: hence "Voigt-Based Fitting". The parameter σ_j of the distribution and the broadening ς of the profile capture, respectively, the *physical* field dispersion of the environment and the *instrumental* broadening; Fitbauer keeps them separate so as not to confuse one with the other.

B.4 Least-squares problem and parametrization

The fit minimizes the residual of (B.4) by nonlinear least squares (`scipy.optimize.least_squares`, trust region) over the parameter vector

$$\boldsymbol{x} \;=\; \big(b,\; s,\; \{A_j,\; \mu_j,\; \log\sigma_j\}_{j=1}^{N},\; \{a_m\}_m\big). \tag{B.7}$$

Details of the parametrization:

- $\log\sigma_j$ **is fitted**, not σ_j: it guarantees $\sigma_j > 0$ without a hard barrier and balances the scale of the width against the position and the area, improving the conditioning.

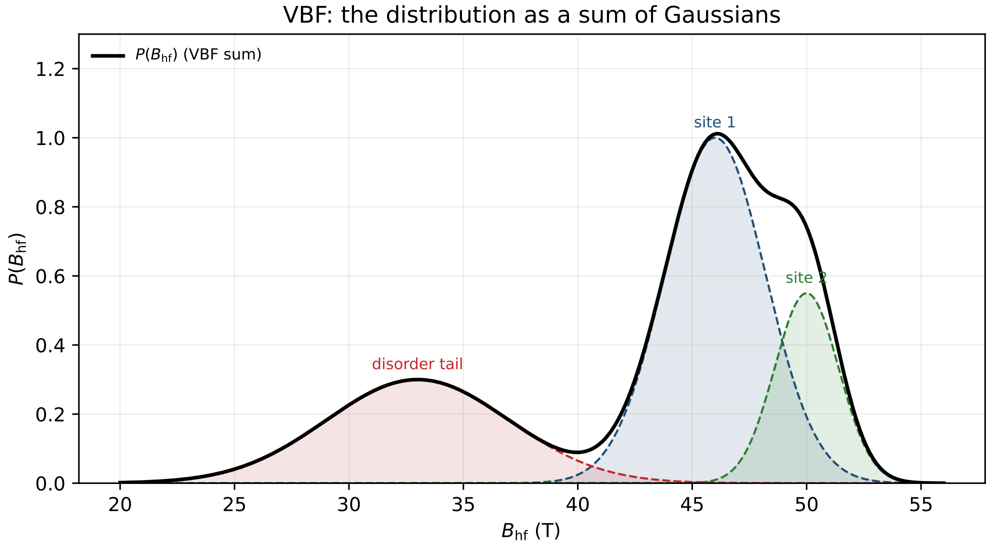

Figure B.1: VBF decomposition (Rancourt–Ping). The distribution $P(B_{\text{hf}})$ is modeled as a sum of Gaussians (B.1); since the Gaussian kernel is normalized to unit area (B.5), the amplitude A_j of each component *is* its area and its fraction $A_j/\sum_i A_i$ gives the population of the environment. The sum reproduces the distribution with few parameters and without regularization.

- **Physical bounds**: $A_j \geq 0$ (non-negative areas), $\mu_j \in [B_{\text{hf,min}}, B_{\text{hf,max}}]$ within the grid, and σ_j bounded between a fraction of a bin and the full range. The baseline satisfies $b \geq 0$ and the sharp amplitudes $a_m \geq 0$.
- **Start**: the N centers are seeded equispaced over the range, with initial width $\sim (B_{\text{hf,max}} - B_{\text{hf,min}})/(3N)$ and area distributed.
- **Ordering and identifiability**: since the model is invariant under permutations of the components (*label switching*), on completion they are **ordered by** μ_j before being saved, so that component 1 is always the one with the lowest field.

Compared with the histogram, here there is *no* regularization term: the functional form (B.1) itself imposes the smoothness, and for this reason the selection of α and the L-curve do not intervene (the corresponding button is disabled).

B.5 Areas, populations and percentages

Since each Gaussian kernel of (B.5) integrates to unity over the grid, the area of component j in $P(B_{\text{hf}})$ is exactly its coefficient:

$$\int A_j \, \mathcal{N}(B_{\text{hf}}; \mu_j, \sigma_j) \, \mathrm{d}B_{\text{hf}} \;=\; A_j. \tag{B.8}$$

Therefore the **population fraction** of environment j —what proportion of the Fe nuclei see that mean field— is, in percent,

$$f_j \;=\; 100 \, \frac{A_j}{\sum_{i=1}^{N} A_i}. \tag{B.9}$$

Fitbauer reports A_j, f_j, μ_j and σ_j per component in the status panel and in the reports. This is the operational advantage of the VBF: it **directly quantifies the proportion** of two or more magnetic subpopulations, something that a histogram only gives after integrating regions of the profile by hand.

B.6 Correlation $\delta(H)$ and $\Delta E_Q(H)$

The VBF admits the same correlation slopes as the histogram (§A.3.9): when building each column k of the kernel (B.2) one uses

$$\delta_k = \delta_0 + \kappa_\delta\,(c_k - H_{\mathrm{ref}}), \qquad \Delta E_{Q,k} = \Delta E_{Q,0} + \kappa_q\,(c_k - H_{\mathrm{ref}}), \tag{B.10}$$

instead of global values (Le Caër–Dubois / Wivel–Mørup model). The model remains linear in the weights w_k; only the kernel changes. With $\kappa_\delta = \kappa_q = 0$ (by default) the classic case is recovered, and the slopes can be fixed or refined in the outer loop.

B.7 Case $N = 1$: the "Gaussian" shape

The **Gaussian** shape of the interface **is the VBF with a single component** ($N = 1$) and a Lorentzian line profile. With this, (B.1) reduces to a single Gaussian of fields,

$$P(B_{\mathrm{hf}}) = A\,\mathcal{N}(B_{\mathrm{hf}}; \mu, \sigma), \tag{B.11}$$

and the parameter vector (B.7) to $(b, s, A, \mu, \log\sigma)$: the minimal description of *one* broadened, approximately symmetric environment. Internally Fitbauer does not maintain a separate algorithm for it —"Gaussian" and "VBF" share exactly the same code— but routes the label to the same fit with $N = 1$. Choose "Gaussian" as a quick first approximation to a single phase; move up to "VBF" with $N = 2$–3 as soon as the residual reveals more than one environment.

B.8 VBF versus the histogram

	Histogram (H.–R.)	**VBF**
Shape parameters	M bin weights ($\sim$40–100)	$3N$ ($N = 1$–6)
Smoothness	imposed by α (regularizer)	imposed by the shape
Selection of α / L-curve	yes, critical	not applicable
Output	free profile	A_j, μ_j, σ_j per environment
Populations	integrating by hand	direct (B.9)
Risk	oscillations if α is wrong	overfitting if N is high

In summary: use the **histogram** when you do not want to commit to a shape (a possibly complex or irregularly multimodal distribution) and the **VBF** when you expect *few well-defined environments* and are interested in their position, width and proportion. Both share the same kernel, the same correlations and the same sharp components.

B.9 Best practices

- **Start low**: $N = 1$–2 and increase only if the residual demands it. With an excessive N, two Gaussians tend to overlap on the same environment and end up poorly determined (anticorrelated areas, degenerate widths).
- **Watch the κ_δ–grid-shift degeneracy**: a slope κ_δ and a global shift of δ are almost indistinguishable; refine with bounds and gently.
- **Separate physics and instrument**: let σ_j absorb the field dispersion and ς (Voigt) the instrumental broadening; do not compensate one with the other.
- **Compare with the histogram**: if the $P(B_{\mathrm{hf}})$ of the regularized histogram and the envelope of the VBF coincide, the description in a few Gaussians is justified.

References

- D. G. Rancourt, J. Y. Ping (1991). *Voigt-based methods for arbitrary-shape static hyperfine parameter distributions in Mössbauer spectroscopy.* Nucl. Instrum. Methods Phys. Res. B **58**, 85–97.
- G. Le Caër, J. M. Dubois (1979). *Evaluation of hyperfine parameter distributions from overlapped Mössbauer spectra of amorphous alloys.* J. Phys. E: Sci. Instrum. **12**, 1083–1090.
- C. Wivel, S. Mørup (1981). *Improved computational procedure for evaluation of overlapping hyperfine parameter distributions in Mössbauer spectra.* J. Phys. E: Sci. Instrum. **14**, 605–610.

Appendix C

Isomer-shift distributions $P(\mathrm{IS})$

C.1 Physical motivation

The isomer shift δ measures the shift of the center of the Mössbauer pattern with respect to a reference, usually α-Fe. It is related to the electron density at the nucleus and, therefore, to the oxidation state, coordination and covalency of the Fe atom. In ordered materials a single δ is usually fitted per phase. However, in glasses, silicates, amorphous materials, ferritins, nanoparticles or mixtures of local environments, a continuous distribution of shifts may exist.

Fitbauer implements two extensions:

- a one-dimensional distribution $P(\delta)$;
- two-dimensional distributions that combine δ with another quantity, especially $P(\delta, \Delta E_Q)$ and $P(B_{\mathrm{hf}}, \delta)$.

The $P(\delta, \Delta E_Q)$ case is particularly useful for broad paramagnetic doublets, where the center of the doublet and its separation may vary simultaneously.

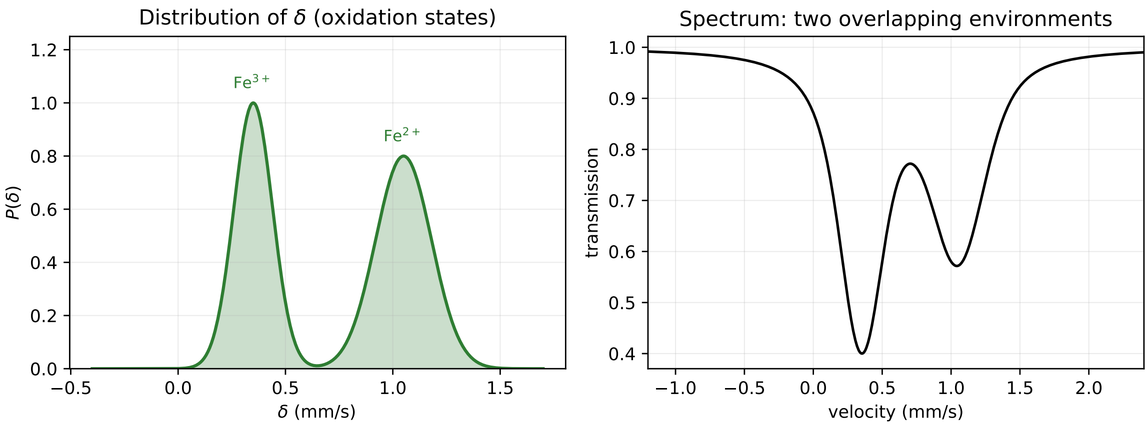

Figure C.1: Isomer shift distribution. *Left*: a $P(\delta)$ with two populations separated in δ —for example Fe^{3+} ($\delta \approx 0{,}35$ mm/s) and Fe^{2+} ($\delta \approx 1{,}05$ mm/s)— reflects two oxidation states or environments. *Right*: the resulting spectrum is the superposition of the patterns of each δ; when the environments overlap, the distribution separates them better than a rigid discrete fit.

C.2 One-dimensional distribution $P(\delta)$

For a grid of shifts δ_i, the 1D model is

$$T(v_k) = b_0 + b_1 v_k - \sum_i P_i\, S(v_k; \delta_i, \Delta E_Q, B_{\mathrm{hf}}, \Gamma), \tag{C.1}$$

where S is the unit-depth absorption pattern. If $B_{\mathrm{hf}} = 0$, the pattern reduces in practice to a doublet or singlet depending on the value of ΔE_Q. The weights satisfy

$$P_i \geq 0. \tag{C.2}$$

The regularized problem is written as

$$\Phi(\boldsymbol{p}) = \|W(\boldsymbol{y} - X\boldsymbol{p})\|^2 + \alpha\|D^2\boldsymbol{p}\|^2, \tag{C.3}$$

where D^2 is the second-difference operator. The normalized probability is computed as

$$p(\delta_i) = \frac{P_i}{\int P(\delta)\,\mathrm{d}\delta}. \tag{C.4}$$

C.3 Two-dimensional distribution $P(\delta, \Delta E_Q)$

For a grid δ_i, Q_j, the model is

$$T(v_k) = b_0 + b_1 v_k - \sum_{ij} P_{ij}\, S(v_k; \delta_i, Q_j, B_{\mathrm{hf}}, \Gamma). \tag{C.5}$$

If one wishes to describe a paramagnetic population, $B_{\mathrm{hf}} = 0$ is fixed and each point (δ_i, Q_j) generates a doublet centered at δ_i with separation Q_j:

$$v_{1,ij} = \delta_i - \frac{Q_j}{2}, \tag{C.6}$$

$$v_{2,ij} = \delta_i + \frac{Q_j}{2}. \tag{C.7}$$

The 2D regularization penalizes curvature in both directions:

$$\Phi(\boldsymbol{P}) = \|W(\boldsymbol{y} - X\boldsymbol{z})\|^2 + \alpha_\delta\|D_\delta^2 P\|^2 + \alpha_Q\|D_Q^2 P\|^2, \qquad P_{ij} \geq 0. \tag{C.8}$$

In augmented form:

$$\begin{bmatrix} WX \\ \sqrt{\alpha_\delta}\, D_\delta^2 \\ \sqrt{\alpha_Q}\, D_Q^2 \end{bmatrix} \boldsymbol{z} \simeq \begin{bmatrix} W\boldsymbol{y} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}. \tag{C.9}$$

C.4 Interpretation of $P(\delta, \Delta E_Q)$

The marginals are

$$P_\delta(\delta_i) = \int P(\delta_i, Q)\,\mathrm{d}Q, \tag{C.10}$$

$$P_Q(Q_j) = \int P(\delta, Q_j)\,\mathrm{d}\delta. \tag{C.11}$$

A tilted cloud in the map may indicate that the sites with higher δ also have a higher or lower ΔE_Q. This may suggest families of local environments, for example variation in valence, coordination or polyhedron distortion.

Fitbauer computes descriptive diagnostics of the map:

$$\langle\delta\rangle, \quad \sigma_\delta, \quad \langle Q\rangle, \quad \sigma_Q, \quad \rho_{\delta Q}. \tag{C.12}$$

The apparent correlation is

$$\rho_{\delta Q} = \frac{\langle(\delta - \langle\delta\rangle)(Q - \langle Q\rangle)\rangle}{\sigma_\delta \sigma_Q}. \tag{C.13}$$

These values describe the fitted distribution, but on their own they do not prove that the distribution is unique or physical.

C.5 Distribution $P(B_{\mathrm{hf}}, \delta)$

$P(B_{\mathrm{hf}}, \delta)$ is also implemented:

$$T(v_k) = b_0 + b_1 v_k - \sum_{ij} P_{ij}\, S(v_k; B_i, \Delta E_Q, \delta_j, \Gamma). \tag{C.14}$$

It can be useful in magnetic phases where the hyperfine field and the isomer shift vary jointly, for example due to a size distribution, partial oxidation or surface environments. However, it is especially sensitive to velocity calibration errors: a global shift of the scale can look like an δ distribution.

C.6 Identifiability risks

Important warning: distributions that contain δ are very sensitive to calibration, folding and background. In a doublet,

$$v_1 = \delta - \frac{\Delta E_Q}{2}, \qquad v_2 = \delta + \frac{\Delta E_Q}{2}, \tag{C.15}$$

so that changes in δ shift the center and changes in ΔE_Q modify the separation. With broad lines or noise, many $(\delta, \Delta E_Q)$ combinations can produce similar spectra. In addition, δ correlates with:

- the folding point;
- the V_{max} calibration;
- the isomeric reference;
- the width Γ;
- the slope/background;
- omitted discrete components.

Therefore, a $P(\delta)$ or $P(\delta, \Delta E_Q)$ distribution that is visually smooth and has a good residual may lack physical meaning if it is compensating for an instrumental error or an incomplete model.

C.7 Good practices

Before interpreting a distribution involving δ it is recommended to:

1. fix or carefully verify the velocity calibration;
2. use a known isomeric reference;
3. check the folding point with the residual diagnostic;
4. compare with discrete models and with the 1D $P(\Delta E_Q)$;
5. vary α_δ and α_Q over several orders of magnitude;
6. repeat with a different number of bins;
7. avoid publishing fine details that disappear when the regularization is changed;
8. use external information: chemistry, XRD, TEM, magnetometry or the literature.

For publication, it is advisable to show that the $P(\delta, \Delta E_Q)$ map is stable and that a simpler model does not explain the data satisfactorily.

Appendix D

Two-dimensional distributions $P(B_{\mathrm{hf}}, \Delta E_Q)$

D.1 Motivation

The one-dimensional distributions $P(B_{\mathrm{hf}})$, $P(\Delta E_Q)$ and $P(\delta)$ are appropriate when the dominant source of disorder is, respectively, magnetic, quadrupolar or isomer-shift related. In amorphous materials, disordered ferrites, glasses, nanoparticles or phases with multiple local environments it may happen that two parameters change at the same time. In that case one can pose a two-dimensional distribution:

$$P(x, y) \geq 0, \qquad x, y \in \{B_{\mathrm{hf}}, \Delta E_Q, \delta\}. \tag{D.1}$$

The cases implemented in the GUI are $P(B_{\mathrm{hf}}, \Delta E_Q)$, $P(\delta, \Delta E_Q)$ and $P(B_{\mathrm{hf}}, \delta)$. The main interest is being able to study not only the marginals, but also the possible correlation between the two distributed quantities. The $P(\delta, \Delta E_Q)$ case is especially relevant for broad paramagnetic doublets in glasses, silicates and amorphous phases.

D.2 Spectral model

Let v_k be the velocity channel and let $S(v; B, Q, \delta, \Gamma)$ be the absorption sextet of unit depth. On a grid x_i, y_j the model that is linear in the weights is

$$T_k = b_0 + b_1 v_k - \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} P_{ij}\, S(v_k; B_{ij}, Q_{ij}, \delta_{ij}, \Gamma). \tag{D.2}$$

The non-distributed parameters are kept global. For example, in $P(\delta, \Delta E_Q)$ a global B is fixed or fitted; if $B = 0$ the pattern represents a set of magnetically collapsed doublets. In matrix form,

$$\boldsymbol{y} \simeq X\boldsymbol{z}, \tag{D.3}$$

where $\boldsymbol{z}$ contains baseline, slope and the flattened weights P_{ij}. The columns associated with the distribution are $-S(v_k; B_i, Q_j)$ because absorption subtracts transmission.

D.3 2D regularization

The inverse problem is strongly underdetermined. For this reason the curvature of the distribution is penalized along both directions:

$$\Phi(\boldsymbol{z}) = \|W(\boldsymbol{y} - X\boldsymbol{z})\|^2 + \alpha_B \|D_B^2 P\|^2 + \alpha_Q \|D_Q^2 P\|^2, \tag{D.4}$$

subject to the constraint

$$P_{ij} \geq 0. \tag{D.5}$$

Here W is the statistical weight of the channels, D_B^2 applies second differences along $B_{\rm hf}$, and D_Q^2 applies second differences along ΔE_Q.

Dimensionless α_B, α_Q. As in 1D, the α values set by the user are not the raw weights: each axis is **non-dimensionalized** independently, $\alpha_{B,\rm eff} = \alpha_B\,\lambda_B\,c_0$ and $\alpha_{Q,\rm eff} = \alpha_Q\,\lambda_Q\,c_0$, with $\lambda = \|A_{\rm dist}\|_F^2/\|L_{\rm eje}\|_F^2$ (Frobenius norm of the whitened operators) and $c_0 = 10^{-4}$ (`ALPHA_REF_SCALE`). Thus $\log_{10}\alpha_B \approx \log_{10}\alpha_Q \approx 0$ is the natural balance on each axis, independent of N, noise and depth. The augmented system uses $\sqrt{\alpha_{\cdot,\rm eff}}\,L_\cdot$. In the initial implementation a rectangular grid is used and the operators are written by means of Kronecker products:

$$L_B = D_B^2 \otimes I_Q, \tag{D.6}$$

$$L_Q = I_B \otimes D_Q^2. \tag{D.7}$$

The augmented problem solved by bounded least squares is

$$\begin{bmatrix} WX \\ \sqrt{\alpha_B}\,L_B \\ \sqrt{\alpha_Q}\,L_Q \end{bmatrix} \boldsymbol{z} \simeq \begin{bmatrix} W\boldsymbol{y} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \qquad P_{ij} \geq 0. \tag{D.8}$$

D.4 Marginals, moments and correlation

Once P_{ij} has been fitted, the marginal distributions are

$$P_B(B_i) = \int P(B_i, Q)\,\mathrm{d}Q \simeq \sum_j P_{ij}\,\Delta Q_j, \tag{D.9}$$

$$P_Q(Q_j) = \int P(B, Q_j)\,\mathrm{d}B \simeq \sum_i P_{ij}\,\Delta B_i. \tag{D.10}$$

On a non-uniform grid, appropriate integration weights must be used. The normalized two-dimensional probability is defined by

$$p_{ij} = \frac{P_{ij}}{\iint P(B,Q)\,\mathrm{d}B\,\mathrm{d}Q}. \tag{D.11}$$

With p_{ij} the diagnostic moments are computed,

$$\langle B\rangle = \sum_{ij} p_{ij} B_i\,\Delta B_i \Delta Q_j, \qquad \langle Q\rangle = \sum_{ij} p_{ij} Q_j\,\Delta B_i \Delta Q_j, \tag{D.12}$$

$$\sigma_B^2 = \sum_{ij} p_{ij}(B_i - \langle B\rangle)^2 \Delta B_i \Delta Q_j, \qquad \sigma_Q^2 = \sum_{ij} p_{ij}(Q_j - \langle Q\rangle)^2 \Delta B_i \Delta Q_j, \tag{D.13}$$

and the apparent correlation

$$\rho_{BQ} = \frac{\sum_{ij} p_{ij}(B_i - \langle B\rangle)(Q_j - \langle Q\rangle)\Delta B_i \Delta Q_j}{\sigma_B \sigma_Q}. \tag{D.14}$$

A tilted cloud in the B–Q map may suggest a correlation between field and quadrupole, but that interpretation is only valid if the solution is stable.

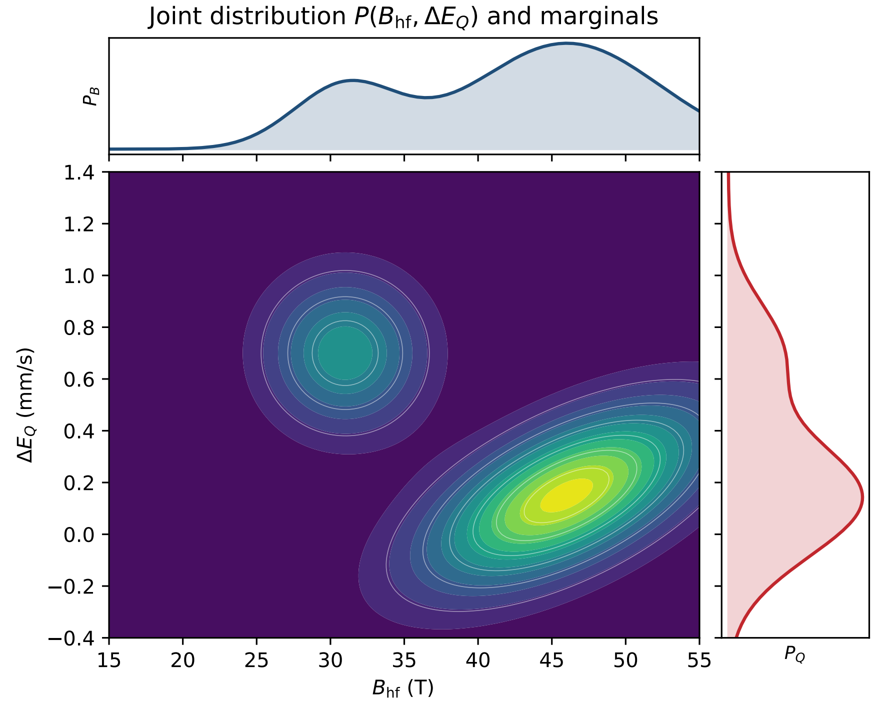

Figure D.1: Joint distribution $P(B_{\mathrm{hf}}, \Delta E_Q)$ and its marginals. The central map shows two populations: a *tilted* cloud (high field, small quadrupole), whose tilt indicates an apparent correlation ρ_{BQ}, and a second isolated site. At the top and to the right, the marginals P_B and P_Q obtained by integrating the map; on their own they do not reveal the correlation, which is only visible in the 2D map.

D.5 Simultaneous sharp components

The implementation allows sharp components to be added to the 2D problem. The model is extended to

$$T_k = b_0 + b_1 v_k - \sum_{ij} P_{ij} S_{ij}(v_k) - \sum_m a_m C_m(v_k) - F_k, \tag{D.15}$$

where C_m are discrete components with free amplitude $a_m \geq 0$ and F_k is the known absorption of components with fixed depth. The sharp columns are not regularized; they compete with the distribution only through the residual. This is useful for separating a known discrete phase from a disordered background, but it can also transfer weight artificially between map and sharp components if they overlap.

D.6 2D regularization L-surface

In 1D an L-curve is used. In 2D there are two regularizations, so the analogue is a surface in (α_B, α_Q). For each pair the following are computed:

$$R(\alpha_B, \alpha_Q) = \|W(\boldsymbol{y} - X\boldsymbol{z})\|, \tag{D.16}$$

$$G(\alpha_B, \alpha_Q) = \sqrt{\|D_B^2 P\|^2 + \|D_Q^2 P\|^2}, \tag{D.17}$$

$$\mathrm{RMS}(\alpha_B, \alpha_Q) = \sqrt{\langle r_k^2 \rangle}. \tag{D.18}$$

The GUI shows a map of RMS versus $\log_{10} \alpha_B$ and $\log_{10} \alpha_Q$ and a projection of R versus G. Two points are proposed: the RMS minimum and a geometric compromise between residual and roughness. Neither is an automatic answer; the stability of the map must be verified visually.

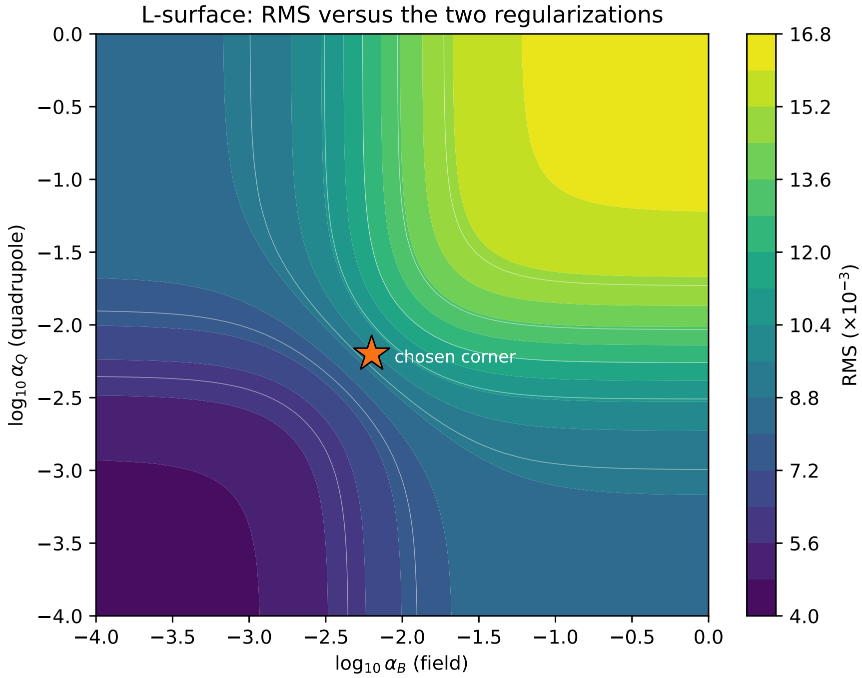

Figure D.2: 2D regularization L-surface, analogous to the L-curve but with two parameters. The RMS is low when both regularizations are weak (lower-left corner: the map fits the noise) and grows as they are strengthened (upper right: over-smoothing). The compromise corner (star) balances fit and smoothness on each axis; it is advisable to check that the solution is stable around it.

D.7 Overfitting risk

Essential warning: a 2D distribution may apparently fit very well and have no physical meaning. For example, a 30×30 grid contains 900 weights, often more than the effective number of independent points in the spectrum. With weak regularization, the map can absorb:

- statistical noise,
- folding or velocity-scale errors,
- poorly chosen line widths,
- instrumental background,
- omitted discrete components,
- dynamic relaxation that should be modeled in another way.

Therefore, a low residual or a visually attractive map do not by themselves prove the existence of a physical distribution $P(B_{\mathrm{hf}}, \Delta E_Q)$.

D.8 Good practices

Before interpreting the map, it is recommended to:

1. compare with simpler models: discrete sextets, $P(B_{\mathrm{hf}})$, $P(\Delta E_Q)$ and sharp components;
2. vary α_B and α_Q over several orders of magnitude;

3. repeat with a different number of bins;
4. verify that the marginals and main features are stable;
5. fix δ and Γ if they are known independently;
6. check that the residual retains no systematic structure;
7. cross-check against external information: TEM, magnetometry, structure, chemistry or literature.

For publication, the 2D model should be justified by showing that a 1D or discrete model does not explain the data satisfactorily and that the 2D distribution is robust against reasonable changes of regularization.

Appendix E

Magnetic relaxation models

This document describes the three magnetic relaxation models implemented in Fitbauer, complementary to the pre-existing singlet, doublet, sextet and Hesse–Rübartsch distribution models: (1) `Relajacion`: a phenomenological component that interpolates between a blocked sextet and an apparent superparamagnetic doublet; (2) `BlumeTjon`: a dynamic two-state model ($+B_{\mathrm{hf}} \leftrightarrow -B_{\mathrm{hf}}$) with symmetric Bloch–McConnell-type exchange; (3) `NeelSize`: a Néel–Arrhenius model with a lognormal particle size distribution, combining the `BlumeTjon` profile for each diameter. Simultaneous global fitting of spectra at several temperatures is available from **Fit → Global Néel (T)**.

E.1 Physical motivation

In magnetic nanoparticles, ferrites or partially superparamagnetic iron oxides, the magnetic moment may fluctuate during the Mössbauer observation window. If the fluctuations are slow, the nucleus observes an almost static hyperfine field and the spectrum is a sextet. If they are fast, the magnetic field is averaged out and a superparamagnetic doublet or singlet appears. In the intermediate regime the lines broaden and coalesce.

The implementation aims to avoid an automatic "sextet + doublet = two chemical phases" interpretation. A doublet-like fraction may be the spectral manifestation of fast relaxation.

> **Important**
>
> The models described here are complementary. If the `Relajacion` or `BlumeTjon` components are not selected, the previous models are evaluated by exactly the same code branches as before.

E.2 Common notation

In the thin-absorber limit, the transmission is written as

$$T(v) = b + s\,v - A(v), \tag{E.1}$$

where b is the baseline, s the slope and $A(v)$ the positive absorption of the component. For a conventional sextet:

$$A_6(v) = D \sum_{j=1}^{6} I_j L\big(v; v_j, \Gamma_j\big), \tag{E.2}$$

with depth D, intensities I_j and Lorentzian profile

$$L(v; v_0, \Gamma) = \frac{\Gamma^2}{(v - v_0)^2 + \Gamma^2}. \tag{E.3}$$

The first-order positions are

$$v_j = \delta + q_j\,\Delta E_Q + m_j\,\frac{B_{\rm hf}}{33\,{\rm T}}, \tag{E.4}$$

where m_j are the α-Fe reference positions at 33 T and q_j the first-order quadrupole pattern.

E.3 Phenomenological model `Relajacion`

E.3.1 Definition

The `Relajacion` component interpolates between a blocked sextet and an apparent superparamagnetic doublet:

$$A_{\rm rel}(v) = D\left[f_{\rm b,ef}\,S_6(v) + \left(1 - f_{\rm b,ef}\right)D_2^*(v)\right]. \tag{E.5}$$

$S_6(v)$ is the unit-depth sextet profile and $D_2^*(v)$ is the doublet scaled so that its integrated area is comparable to that of the sextet:

$$D_2^*(v) = D_2(v)\,\frac{\int S_6(v)\,dv}{\int D_2(v)\,dv}. \tag{E.6}$$

In this way, D approximately retains the meaning of total area/depth.

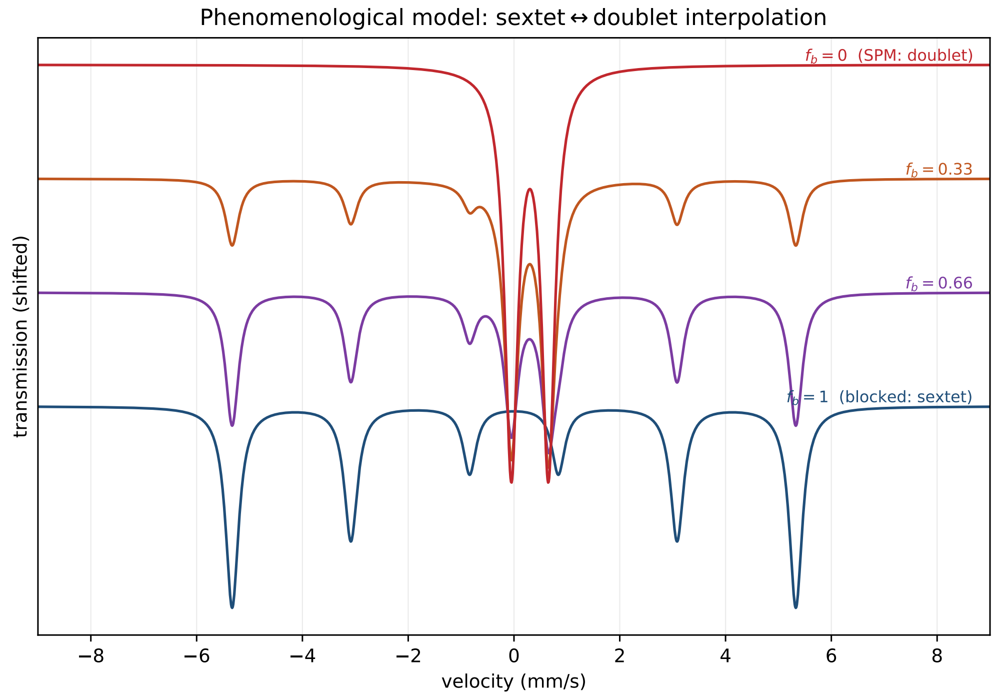

Figure E.1: Phenomenological model `Relajacion` (E.5). As the blocked fraction f_b decreases from 1 (blocked sextet) to 0 (apparent superparamagnetic doublet), the sextet signal is transferred to a central doublet of equivalent area. It does not represent two chemical phases, but the spectral effect of relaxation.

E.3.2 Blocked fraction and phenomenological rate

The user sees two specific parameters:

- $f_{\rm bloq}$: maximum blocked fraction.
- $\log_{10}\nu$: phenomenological relaxation rate in ${\rm s}^{-1}$.

The fraction that actually enters (E.5) is

$$f_{\mathrm{b,ef}} = f_{\mathrm{bloq}}\, F(\log_{10}\nu), \tag{E.7}$$

with

$$F(x) = \frac{1}{1+\exp\left(\frac{x-x_c}{w}\right)}, \qquad x_c = 8.5, \quad w = 0.55. \tag{E.8}$$

Thus, slow rates ($x \ll x_c$) leave $F \simeq 1$, whereas fast rates ($x \gg x_c$) force $F \simeq 0$.

In addition, a maximum phenomenological broadening is applied in the intermediate regime:

$$\Gamma_{\mathrm{ef}} = \Gamma\left[1 + 1.6\exp\left(-\frac{1}{2}\left(\frac{x-x_c}{0.75}\right)^2\right)\right]. \tag{E.9}$$

Important correlation

f_{bloq} and $\log_{10}\nu$ are not fully independent parameters: both reduce the observed sextet fraction. In practice, it is advisable to fix one of the two unless the spectrum carries enough information. To estimate fractions, fix $\log_{10}\nu$ and free f_{bloq}; to study dynamics, fix $f_{\mathrm{bloq}} = 1$ and free $\log_{10}\nu$.

E.4 Dynamic model `BlumeTjon`: two states

E.4.1 Exchange matrix

The `BlumeTjon` component implements a first dynamic version with two opposite field orientations:

$$+B_{\mathrm{hf}} \leftrightarrow -B_{\mathrm{hf}}, \tag{E.10}$$

with symmetric transition matrix

$$W = \begin{pmatrix} -\nu & \nu \\ \nu & -\nu \end{pmatrix}. \tag{E.11}$$

There is no explicit blocked fraction: the dynamics is controlled by ν.

E.4.2 Exchange profile for one transition

For each sextet line, two frequencies/velocities are computed, one for $+B_{\mathrm{hf}}$ and one for $-B_{\mathrm{hf}}$, denoted v_a and v_b. A symmetric exchange form of Bloch–McConnell/Kubo type is used. We define

$$z_a = \Gamma + i(v - v_a), \qquad z_b = \Gamma + i(v - v_b), \tag{E.12}$$

and an exchange rate in velocity units

$$k_v = \frac{\nu}{\mathcal{C}_{\nu\to v}}, \tag{E.13}$$

where

$$\mathcal{C}_{\nu\to v} = \frac{E_\gamma}{h c_{\mathrm{mm/s}}}. \tag{E.14}$$

Here E_γ is the energy of the ^{57}Fe transition and $c_{\mathrm{mm/s}}$ the speed of light in mm/s. The absorption profile used is the real part of

$$R(v) = \frac{1}{2}\Gamma\,\frac{z_a + z_b + 4k_v}{z_a z_b + k_v(z_a + z_b)}. \tag{E.15}$$

The total contribution is

$$A_{\mathrm{BT}}(v) = D\sum_{j=1}^{6} I_j\, \Re\left[R_j(v)\right]_+, \tag{E.16}$$

where $[\cdot]_+$ clips small numerical negative contributions.

E.4.3 Limits

- **Slow limit**: $k_v \to 0$. Equation (E.15) tends to the average of two static Lorentzians. In the symmetric $+B_{\mathrm{hf}}/-B_{\mathrm{hf}}$ pattern it reproduces the blocked sextet.
- **Intermediate**: k_v comparable to the line separation. Broadening and coalescence appear.
- **Fast**: large k_v. The magnetic positions are averaged out and the spectrum tends to a doublet or singlet depending on ΔE_Q.

Figure E.2 shows the continuous evolution between these limits as the exchange rate increases.

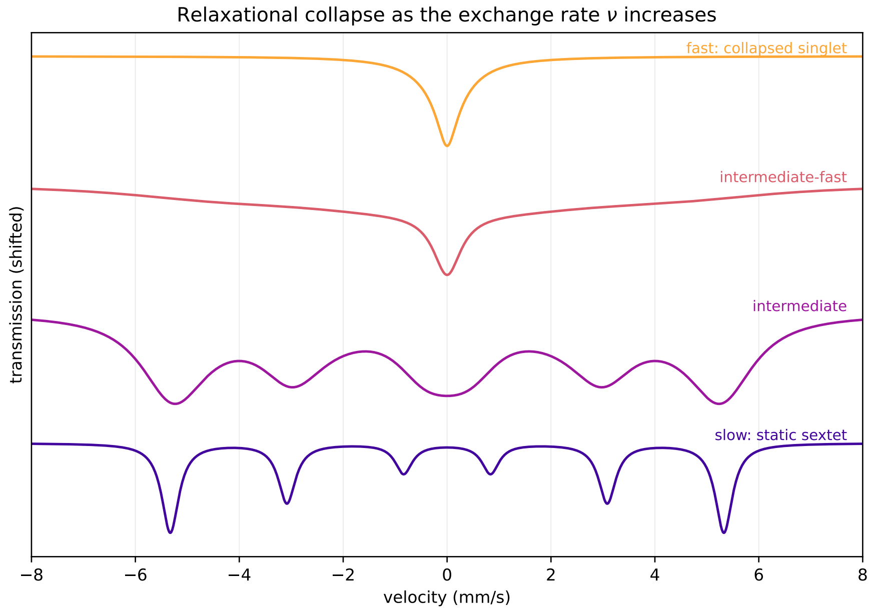

Figure E.2: Relaxation-induced collapse in the two-state model $+B_{\mathrm{hf}} \leftrightarrow -B_{\mathrm{hf}}$ (E.15). At a low rate (bottom) the spectrum is the *static sextet*; as the rate grows the lines broaden and coalesce (*intermediate regime*); at a high rate (top) the magnetic positions are averaged to zero and a *single collapsed line* remains. Curves offset vertically for comparison.

E.5 Néel–Arrhenius and size distribution

The `NeelSize` component uses the Néel–Arrhenius model to convert the particle volume into a relaxation rate. For spherical particles of diameter d,

$$V(d) = \frac{\pi d^3}{6}, \tag{E.17}$$

and

$$\tau_N(d,T) = \tau_0 \exp\left(\frac{K_{\mathrm{eff}} V(d)}{k_B T}\right), \qquad \nu(d,T) = \tau_0^{-1} \exp\left(-\frac{K_{\mathrm{eff}} V(d)}{k_B T}\right). \tag{E.18}$$

In the GUI, $\log_{10} K_{\mathrm{eff}}$ and $\log_{10} \tau_0$ are fitted to keep reasonable numerical ranges:

$$\log_{10} \nu_i = -\log_{10} \tau_0 - \frac{K_{\mathrm{eff}} V(d_i)}{k_B T \ln 10}. \tag{E.19}$$

The size distribution is initially modelled as lognormal. If d_{50} is the median and σ_d the lognormal width,

$$f(d) = \frac{1}{d\sigma_d\sqrt{2\pi}} \exp\left[-\frac{(\ln d - \ln d_{50})^2}{2\sigma_d^2}\right]. \tag{E.20}$$

The total spectrum is discretized as

$$A_{\mathrm{Neel}}(v) = D \sum_i w_i \, A_{\mathrm{BT}}\big(v; \log_{10} \nu_i, T, K_{\mathrm{eff}}, d_i\big), \tag{E.21}$$

where A_{BT} is the two-state exchange profile of the previous section. Large particles have a larger barrier $K_{\mathrm{eff}} V$ and therefore relax more slowly; small ones relax fast and contribute a superparamagnetic signal.

The approximate blocking temperature for a given diameter is estimated with

$$T_B(d) = \frac{K_{\mathrm{eff}} V(d)}{k_B \ln(\tau_M/\tau_0)}, \tag{E.22}$$

where τ_M is the Mössbauer observation time (by default 10^{-8} s is used for the GUI summary).

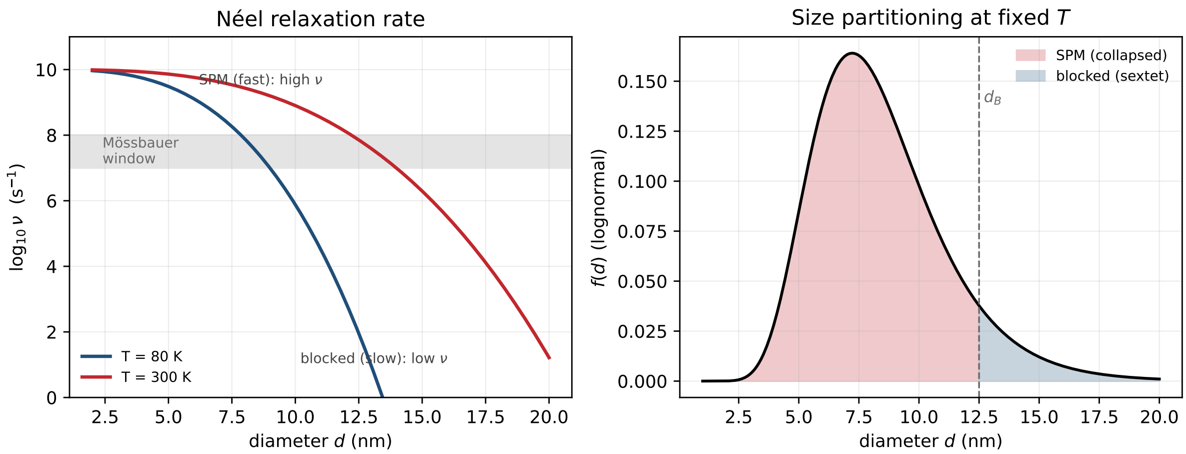

Figure E.3: Néel–Arrhenius: size-dependent blocking. *Left*: the relaxation rate $\nu(d, T)$ drops as the diameter grows; small particles relax fast (superparamagnetic, collapsing to a singlet/doublet) and large ones remain *blocked* (sextet). A particle appears blocked when ν falls below the Mössbauer observation window (grey band); as T decreases the blocking shifts to smaller diameters. *Right*: over a lognormal size distribution, at fixed T the blocking diameter d_B separates the superparamagnetic fraction from the blocked one, and the spectrum is the mixture of both.

Identifiability

In a single spectrum, K_{eff}, d_{50}, σ_d, τ_0, Γ and a possible static distribution $P(B_{\mathrm{hf}})$ may be strongly correlated. It is therefore advisable to fix parameters using external information (TEM, magnetometry, literature) or to use a global multi-temperature fit.

E.6 Global multi-temperature fit

Fitbauer can fit several spectra at different temperatures simultaneously. For each spectrum r there is a $T_r(v)$ and a temperature T_r. The model shares the physical parameters

$$\theta_{\mathrm{shared}} = \{\delta, \Delta E_Q, B_{\mathrm{hf}}, \Gamma, K_{\mathrm{eff}}, d_{50}, \sigma_d, \tau_0\}, \tag{E.23}$$

and keeps local parameters per spectrum, by default

$$\theta_r = \{b_r, s_r, D_r\}. \tag{E.24}$$

The objective function is

$$\chi^2 = \sum_r \sum_k \left(\frac{T_{r,\mathrm{model}}(v_k; \theta_{\mathrm{shared}}, \theta_r, T_r) - T_{r,\mathrm{data}}(v_k)}{\sigma_{rk}} \right)^2 . \tag{E.25}$$

This reduces the degeneracy compared to fitting a single spectrum: the same size distribution must simultaneously explain the blocking at low T and the superparamagnetic collapse at high T.

E.7 Relation to fitting and reports

The fittable parameters are the same as in a sextet (δ, ΔE_Q, B_{hf}, widths, depth and intensities) plus $\log_{10} \nu$ and, only in the phenomenological model, f_{bloq}. In `NeelSize`, T, $\log_{10} K_{\mathrm{eff}}$, d_{50}, σ_d, $\log_{10} \tau_0$ and the number of bins are added. The GUI also shows

$$\tau = \frac{1}{\nu} = 10^{-\log_{10} \nu}\ \mathrm{s} \tag{E.26}$$

and, for `NeelSize`, an approximate T_B.

In reports, the `Relajacion` component should be interpreted as “apparent blocked / superparamagnetic fraction”, not as separate chemical phases. The `BlumeTjon` component is reported as a dynamic two-state model and `NeelSize` as a Néel–Arrhenius model with a size distribution.

Appendix F

Thickness correction

The standard fitting model treats the sample as a *thin* absorber: the transmission is linear in the absorption of each component. In thick or strongly absorbing samples this approximation fails —the deep lines *saturate*— and it biases depths, widths and, above all, the areas (phase fractions). The experimental branch adds a *thickness correction* through a saturation model with a single physical parameter C. This document explains what benefits it brings, when it is worth applying, how it is applied (what changes in the fit and in the display) and how it behaves with several subspectra.

F.1 The problem: thin vs. thick absorber

The historical model is linear in the absorption:

$$T_{\text{delgado}}(v) \;=\; b + s\,v \;-\; A_{\text{tot}}(v), \qquad A_{\text{tot}}(v) = \sum_c A_c(v), \tag{F.1}$$

with b the baseline, s the slope and A_c the absorption of each component (sextet/doublet/singlet). It is exact only in the limit of an infinitely thin absorber. In a real sample of finite thickness, the fraction of resonant photons that reach the detector decays *exponentially* with the amount of resonant material: the deepest lines absorb proportionally less than (F.1) predicts (they *saturate*) and, in addition, they *broaden*.

Note

The thickness saturation effect is not a numerical artifact: it is physics of the transmission experiment. Ignoring it in thick samples makes the fit falsely "compensate" by raising widths and unbalancing areas.

F.2 Exponential saturation model

The correction replaces (F.1) with

$$\boxed{T(v) \;=\; b + s\,v \;-\; C\left(1 - e^{-A_{\text{tot}}(v)/C}\right)} \tag{F.2}$$

where $C > 0$ is the *saturation scale* (the maximum achievable absorption amplitude, $\approx f_s\,b$, with f_s the effective recoil-free fraction of the source). Its two limits make it interpretable and backward compatible:

- **Thin limit** $C \to \infty$: since $C(1 - e^{-x/C}) \to x$, one recovers (F.1) exactly. That is, the thin model is the particular case of zero thickness.

- **Saturation** C finite: the effective absorption $A_{\text{eff}} = C(1 - e^{-A_{\text{tot}}/C})$ is bounded by C and compresses the deep lines the more, the larger A_{tot}/C is.

Implemented in `core/physics.py` (`total_model`, parameter `sat_scale`$=C$).

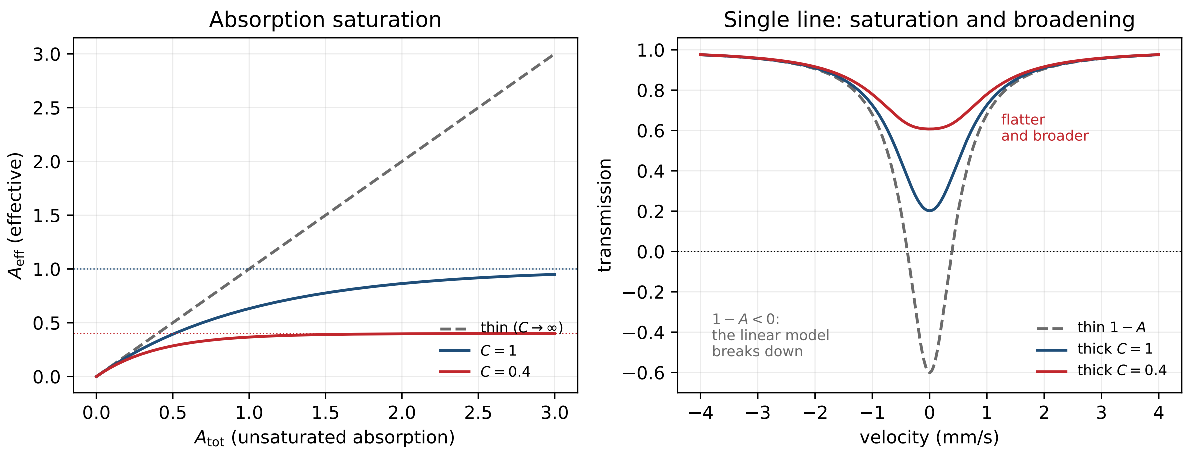

Figure F.1: Thin versus thick absorber. *Left*: the effective absorption $A_{\text{eff}} = C(1 - e^{-A_{\text{tot}}/C})$ saturates, bounded by C, while the thin model ($C \to \infty$) grows without limit. *Right*: over a deep line, the thin model $1 - A$ would reach impossible negative values; the thickness correction gives *shallower and broader* lines, which is the physical effect of saturation.

F.3 Benefits

1. **Unbiased areas and phase fractions.** Saturation suppresses the absorption of the majority phases (deep lines) more than that of the minority ones. Without correction, the area ratio is biased in favor of the weak phases. By *de-saturating*, the recovered areas correspond to those of the equivalent thin absorber, which are the ones proportional to the abundances (at equal Lamb–Mössbauer f).
2. **Correct depths.** The depth parameter no longer reads the apparent (clipped) absorption but represents the real absorption.
3. **Less inflated widths.** The thickness broadening is no longer artificially absorbed into γ; the fitted widths approach the intrinsic ones.
4. **Better χ^2 in deep spectra.** Where the thin model leaves systematic structure in the residual (at the minima), saturation removes it without adding spurious components.

Verification

On a synthetic spectrum saturated with $A_{\max}/C = 2$, the fit with thickness correction reconstructed the spectrum $\sim 27\%$ better (RMS) than the thin model, and recovered the starting scale C ($C = 0.41$ versus the true 0.40). In mild saturation it recovered $\sim 9\%$ more area (de-saturation).

F.4 When to apply it

Important

The thickness correction **helps** when the sample is thick / strongly absorbing and **is not necessary** (and should not be forced) when it is thin.

Signs that it is worth enabling it:

- Deep absorption: the transmission minimum drops well below the baseline (typically $\gtrsim$ 15–20% absorption), or effective thickness $t_a \gtrsim 1$.
- The thin fit leaves *structured residual at the minima* (the model does not reach the bottom of the deep lines or makes them too wide).
- Reliable *phase quantification* is sought (area ratios), not just positions.

When *not*:

- Thin spectra (absorption of a few %): the linear model is already exact and C is undetermined (it tends to large values). Keep thin mode.
- As a diagnostic: if, when C is fitted, it converges to a large value and the χ^2 does not improve, the sample is effectively thin.

F.5 How it is applied in the fit

F.5.1 Selection and parameter

In the calibration panel there is a selector *Absorber:* ***thin*** */* ***thickness*** and a saturation-scale C *slider* (`sat_scale`), active only in thick mode. C is a global parameter (like the baseline or the slope), adjustable or fixable.

F.5.2 Discrete mode

The nonlinear optimizer (TRF) already fits an arbitrary function, so it is enough for the forward model to use (F.2): C enters as one more free parameter and is refined together with the rest. The algorithm does not change, only the forward model.

F.5.3 Distribution mode $P(B_{\mathrm{hf}})$

Here the exponential *breaks the linearity* in the weights P, on which the whole regularized solver relies. The solution is an *inverse transform* that recovers linearity: given the datum y, the saturation is inverted to obtain the observed linear absorption

$$A_{\mathrm{obs}}(v) \;=\; -C\,\ln\!\Big(1-\tfrac{b+s\,v-y(v)}{C}\Big), \tag{F.3}$$

which *is* linear in P ($A_{\mathrm{obs}} = K\,P + \ldots$). The usual regularized linear system (Tikhonov/TV, non-negativity) is solved over A_{obs}, and the model is *re-saturated* with (F.2) to compute residuals and the displayed curve. The noise propagates through the transform,

$$\sigma_A \;=\; \sigma_y\,\frac{dA_{\mathrm{obs}}}{dy} \;=\; \sigma_y\,e^{A_{\mathrm{obs}}/C}, \tag{F.4}$$

so that the saturated channels are weighted correctly. An outer nonlinear loop (separable VARPRO-type scheme) refines (b, s, C) while the inner linear solve recovers P at each iteration.

> **Note**
>
> The transform (F.3) requires $b+s\,v-y < C$ in every channel (the apparent absorption cannot exceed the saturation plateau); the program clips the argument to the valid domain $(0, 1]$ before the logarithm.

F.6 What changes in the view

- The displayed **model curve** is the *saturated* transmission (F.2): it is the one compared with the data and the one that generates the residual.
- The displayed and exported **depths/areas** are those of the equivalent thin absorber (de-saturated): for this reason, when the correction is enabled, the relative areas may change with respect to the thin fit —which is precisely the intended correction.
- The fitted value of C (saturation scale) appears among the global parameters. A large C indicates a practically thin sample.
- In distribution mode, $P(B_{\mathrm{hf}})$ is shown already de-saturated; its integrated area reflects the real fraction.

F.7 Several subspectra

Saturation is a property of the *whole absorber*, not of each phase separately: the photons of a given velocity pass through all the material and "see" the sum of all the absorptions at that velocity. That is why (F.2) saturates the **total absorption** $A_{\mathrm{tot}} = \sum_c A_c$ with a *single* scale C common to all subspectra, and not each component on its own:

$$T(v) = b + s\,v - C\Big(1 - \exp\big[-\tfrac{1}{C}\textstyle\sum_c A_c(v)\big]\Big). \tag{F.5}$$

Practical consequences:

- **Correct physical coupling.** Where two subspectra overlap, the saturation is larger (the local absorption is the sum); the model captures it automatically.
- **Phase quantification.** After de-saturating, the area ratios between subspectra recover their unbiased value, which is the goal in phase mixtures (e.g. several Fe environments).
- **A single C.** A scale per component is not fitted (it would be unphysical and would degenerate with the depths): C is global and describes the effective thickness of the sample.
- **Compatibility.** In distribution mode, the added "sharp" sextets and the distribution itself share the same correction, because they all contribute to A_{tot}.

Model limitation

It is an effective exponential saturation (Lambert–Beer), which captures the dominant thickness effect (compression, broadening, unbiased areas) with a single parameter. It is not the full Margulies–Ehrman transmission integral (convolution with the source line), which would be a second-order refinement rarely necessary in practice.

F.8 Summary

Aspect	Thickness correction
Model	$T = b + sv - C(1 - e^{-A_{\mathrm{tot}}/C})$
New parameter	C (saturation scale), global, $C \to \infty$ = thin
When	thick samples / deep absorption / phase quantification
Discrete	C free in the TRF; only the forward model changes
Distribution	inverse transform $A_{\mathrm{obs}} = -C\ln(1 - (b + sv - y)/C)$ + VARPRO (b, s, C)
View	saturated curve; areas/P de-saturated; C is shown
Subspectra	a single C saturates the sum $\sum_c A_c$ (physical coupling)
Key benefit	unbiased areas/phase fractions, better widths and χ^2